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Prescriptive Analytics for Multi-Objective Inventory Replenishment: A Decision Support Dashboard for San San Marketing

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Chapter 1 : Introduction

The distribution and warehousing industry continues to face increasing difficulty in operations due to rising client demands and the need for efficient inventory handling across multiple stages in the supply chain process. As businesses expand their operations, maintaining accurate inventory levels becomes more challenging, particularly in balancing storage capacity with timely product availability. Manual inventory management approaches often have difficulty adapting to unpredictable demand patterns, which may lead to overstocking, stockouts, and higher operational costs.

To address these challenges, the proposed capstone project will focus on the development of a Decision Support Dashboard (DSD) that integrates descriptive, predictive, and prescriptive analytics to improve inventory management for a distribution and warehousing business. The proposed DSD will utilize historical sales and inventory data to generate demand forecasts using appropriate time series or machine learning models, which will be selected during the design phase. Additionally, optimization techniques will be integrated to support effective stock level decisions and inventory management.

The integrated approach will explore multi-criteria inventory classification techniques, such as ABC analysis, based on financial consequences and demand characteristics. Product segmentation processes systematize the control of warehouse resources through the identification of high-priority SKUs requiring thorough monitoring via forecasting engines (Zhao, 2024; Zietsman, 2023). The classification model will become a strategic filter as it will direct computational energy toward fast-moving

product groups while simultaneously helping to determine slow-moving products, preventing either operational waste or lost interest (Adilla & Utama, 2025). A shift from generalized management practices to a prioritized classification method will facilitate more efficient distribution processes in terms of balancing the trade-offs between holding cost and customer service level. Ultimately, improved inventory planning and decision-making can be achieved through a data-driven and intelligent approach to demand forecasting and inventory control in a warehousing and distribution environment.

Defined in the following sections are the proposed capstone project's primary goals and objectives, outlining the organization and how operations are managed to facilitate the successful completion of the proposed capstone project. The chapter concludes with the definition of the proposed capstone project's scope, constraints, and an abstract model that will be created based on the information detailed above, along with the challenges or opportunities that will be addressed within the proposed capstone project.

1.1 Definition of Terms

The following are definitions of terms that will be used throughout this paper to clarify the concept of this proposed capstone project as it relates to supply chain management and inventory management.

a) Bad Order (BO)

Any item within the inventory that is either no longer sellable because it has spoiled or expired, or has been rendered unsellable because it has been damaged physically. In this study, the "Bad Order" metric will constitute the majority of the determination of physical waste.

b) Good Order (GO)

A confirmed sales transaction in which goods were successfully sold and delivered to a retail customer, recorded per SKU in SSM's sales register. Good Order logs are the primary source of historical demand data used for forecasting.

c) Stock Keeping Unit (SKU)

A unique alphanumeric code assigned to each distinct product variant managed by SSM, used to track individual products across all transaction records including sales, inventory, purchases, and returns.

d) Decision Support Dashboard (DSD)

A web-application tool built for San San Marketing that helps the procurement team decide when and how much to reorder. It shows demand forecasts, reorder recommendations, and inventory data in one place.

e) Holding Costs

The total costs incurred by the company for products that are not sold and are in storage, including but not limited to: warehouse costs, insurance, opportunity cost of capital tied up in the inventory that is immobilized and not

producing a return.

f) Linear Programming (LP)

A mathematical optimization technique that will be used within the dashboard to determine the quantity of inventory to accept per SKU from each supplier-scheduled allocation. Rather than functioning as a standalone single-objective cost minimizer, the LP serves as the computational backbone of the multi-objective model by operating on a weighted composite objective function.

g) Multi-Objective Optimization

A mathematical approach to decision-making that explicitly balances two or more conflicting objectives within a single model. In this proposed capstone project, the three competing objectives are: (1) minimizing holding costs incurred by excess inventory remaining in the warehouse, (2) minimizing stockout penalty costs arising from insufficient stock to meet forecasted demand, and (3) minimizing Bad Order costs resulting from units that become unsellable due to physical damage or expiration while in storage

h) Reorder Point (ROP)

An inventory level at which a new purchase order will need to be placed in order for the replenishment to arrive before the stockout occurs.

i) Replenishment

The process of restoring inventory levels by ordering additional stock from principal suppliers once existing stock has been depleted to a predetermined threshold.

j) Safety Stock

A buffer quantity of inventory maintained above the expected demand level to protect against uncertainties such as demand variability, supplier lead time delays, and forecast errors.

1.2 Company Background

San San Marketing (SSM) is a regional distribution and logistics company located in Barangay Pinmaludpud, Urdaneta City, Pangasinan, founded by Mr. Alexander Gacutan. The company operates within the fast-moving consumer goods (FMCG) sector and provides services such as warehousing, inventory management, and product distribution. Through these operations, SSM supports the continuous movement of essential goods, including food products, groceries, and sanitary items. As a distribution intermediary, the company plays a significant role in connecting manufacturers to retailers across Northern Luzon.

From an operational standpoint, SSM focuses on maintaining efficient inventory control and ensuring the timely and accurate delivery of products. These functions are important in minimizing stock shortages and overstocking, while also improving the reliability of service provided to its clients. The company serves various customers, including supermarkets, retail stores, and other commercial establishments. By carrying out these responsibilities effectively, SSM contributes to improved supply

chain performance and ensures the consistent availability of products in the market.

In addition, SSM manages a structured supply chain network that covers Pangasinan, La Union, and other areas within the Ilocos Region. SSM's role in linking suppliers, wholesalers, and retailers allows for better coordination in the distribution process. The company helps reduce delivery delays, manage operational costs, and respond more efficiently to customer demands. As a result, SSM supports the smooth flow of goods across different locations, which is essential for maintaining reliable retail operations in the region.

In terms of business significance, SSM plays an important role in enhancing the efficiency and stability of the regional supply chain. SSM operations allow businesses of different sizes to maintain a steady supply of products, which is necessary for meeting consumer demand and sustaining daily operations. As the demand for consumer goods continues to grow in Northern Luzon, the company's logistics and distribution services become increasingly important in supporting business expansion and market growth. Overall, SSM contributes to the improvement of trade efficiency and the development of the local commercial sector.

1.3 Project Context

Operating as a major regional distributor in the Fast-Moving Consumer Goods (FMCG) sector means SSM handles thousands of Stock Keeping Units (SKUs) from suppliers such as Unilever Philippines and Del Monte Philippines. Unilever mainly supplies personal care products, while Del Monte provides food products, particularly ingredients used for cooking pasta and other meals. At present, SSM has access to

years of daily transaction logs. Right now, the purchasing department ignores its historical data. Instead, they rely on an annual inventory estimate provided by the manufacturer, divided evenly across 52 weeks, one of the requirements they must follow. This assumption-based approach forces the company to order goods blindly. Staff cannot accurately account for fluctuating product velocities, sudden holiday sales spikes, or delivery delays from suppliers.

Relying on rigid allocation overlooks real demand patterns and forces purchasing decisions to follow a fixed weekly quota rather than actual consumption During initial discussion with SSM, the firm's representative identified the following practices as the source of three key operational concerns:

1. **Inaccurate Order Planning:** Treating the manufacturer's annual inventory estimate as a fixed weekly allocation assumes all products move uniformly. This overlooks wide variations in item-level demand, resulting in distorted replenishment cycles and frequent mismatches between actual sales velocity and the imposed weekly targets.
2. **Trapped Financial Capital:** The fixed weekly allocation encourages over-ordering of slow-moving items, tying up roughly ₱50M to ₱60M in inventory that sits idle, inflating holding costs and restricting capital that could be used for more strategic investments.
3. **Increased Bad Orders (BO):** Misaligned ordering results in excess cases arriving out of sync with store or warehouse capacity. This increases handling errors, returns, and classification of goods as Bad Orders (BO) due to damage,

mis-picks, or prolonged storage turnover issues, an operational loss caused by inaccurate planning.

Continuing to rely on manual estimation is no longer viable for the company. A strategic and practical alternative is to develop a data-driven Decision Support Dashboard. To accurately anticipate local consumer demand, the proposed analytical approach engine will analyze historical sales data using time-series forecasting techniques to detect cyclical and seasonal patterns. Furthermore, instead of relying on a fixed weekly allocation, the backend computational architecture will apply mathematical optimization algorithms to calculate precise replenishment parameters. These mathematical models aim to achieve an optimal balance between minimizing warehouse holding costs and preventing issues such as stockouts. Ultimately, applying a predictive and prescriptive approach completely removes traditional guesswork, transforming SSM's procurement process into a resilient, optimized, and data-backed operation.

1.4 Purpose and Objectives

The section will provide goal statements and the specific objectives of the DSD. The purpose of this proposed capstone project is to create a DSD for San San Marketing that will assist them with sales forecasting based on historical data and optimize inventory through multi-objective optimization techniques to support tactical purchasing decisions.

1.4.1 General Objective

The primary objective of the proposed capstone project is to develop a DSD for San San Marketing. It will apply predictive modeling and mathematical

optimization techniques to analyze historical sell-out data and generate optimized recommendations for allocation acceptance quantities per SKU. This approach aims to reduce holding costs associated with inactive stock and minimize product waste, in alignment with United Nations Sustainable Development Goal (SDG) 12 on responsible consumption and production.

1.4.2 Specific Objectives

The following specific objectives serve as the requirements to achieve the general objective:

1. To analyze historical sales and inventory data of San San Marketing spanning five (5) years of transaction records to establish baseline performance metrics at the SKU level.
2. To optimize demand forecasting accuracy through the implementation of the appropriate time series models.
3. To formulate an optimized model that will help minimize the Total Inventory Cost.
4. To calculate optimal reorder points and safety stock levels using model-generated demand forecasts, thereby reducing storage costs relative to the existing business process.
5. To develop a Decision Support Dashboard that consolidates relevant descriptive, predictive, and prescriptive models into a unified interface.

1.5 Scope and Project Constraints

This section outlines the strategic focus, analytical boundaries, and operational constraints of the proposed capstone project. It defines the specific data parameters

and modeling frameworks that constitute the Decision Support Dashboard for San San Marketing, as well as the procedural and technical limitations that bound the study.

1.5.1 Project Scope

The primary focus of the proposed capstone project is the development of an integrated Decision Support Dashboard (DSD) designed to optimize inventory management for SSM. The dashboard will consolidate SKU-level stock movements from existing Microsoft Excel records, Stock on Hand snapshots, Bad Order summaries, and supplier annual allocation schedules. The dashboard will serve as the central interface through which inventory classification, demand forecasting, and replenishment acceptance recommendations are generated and monitored.

The dataset will draw from SSM's historical transaction records, with a target coverage of at least five (5) years of stock movement logs. This data depth is intended to ensure that the forecasting components of the dashboard can account for long-term seasonality, cyclical patterns, and structural shifts in demand behavior over time. The specific forecasting methods to be implemented will be determined through a comparative evaluation of candidate models, including but not limited to SARIMA, based on their performance against SSM's actual demand patterns. Similarly, the inventory classification approach will be informed by the analytical frameworks reviewed in the preceding section, with the final methodology selected according to the characteristics of SSM's SKU assortment and the data available.

At its core, the dashboard is designed to evaluate the alignment between SSM's supplier-provided annual allocation schedule and forecasted product demand. SSM operates under a manufacturer-driven replenishment model wherein principal suppliers automatically deliver stock according to a pre-established annual schedule divided into periodic tranches. While the exact classification thresholds and optimization logic will be refined during the development phase, the dashboard is broadly oriented toward directing the greatest analytical and computational resources toward the items that carry the highest operational and financial significance for SSM. This ensures that the dashboard remains both practically actionable and scalable as SSM's inventory profile evolves.

1.5.2 Project Constraints

While the proposed dashboard and predictive models are designed to provide decision support and inventory optimization, the following limitations also apply to them:

- The accuracy and reliability of the generated insights will depend on the quality, consistency, and completeness of San San Marketing’s internal datasets spanning 2020-2025. Significant gaps in historical records may require approximation methods during the data preparation phase.
- The proposed dashboard will not incorporate independent purchase order generation. The prescriptive module evaluates the alignment between supplier-provided weekly allocations and time series-generated demand forecasts.

- The models will primarily use SSM's internal data but with the flexibility to use relevant external data such as but not limited to supplier production capacity and Warehouse constraints.
- The proposed capstone project’s primary focus is to provide decision-support tools only; it will not automate processes or business workflows such as automated supplier order placement, customer relationship management (CRM) activities, or payment processing.
- Processing forecasts for a large number of SKUs may be limited by the hardware available at SSM
- The prescriptive module applies a multi-objective optimization model to determine the optimal quantity to accept from each supplier-scheduled allocation per SKU.
- The scope of the predictive and prescriptive modules is limited to Category A items, and the proposed capstone project’s processing efforts focus on the highest-value products with the greatest impact on financial and operational performance.

1.6 Conceptual Framework of Data Integration

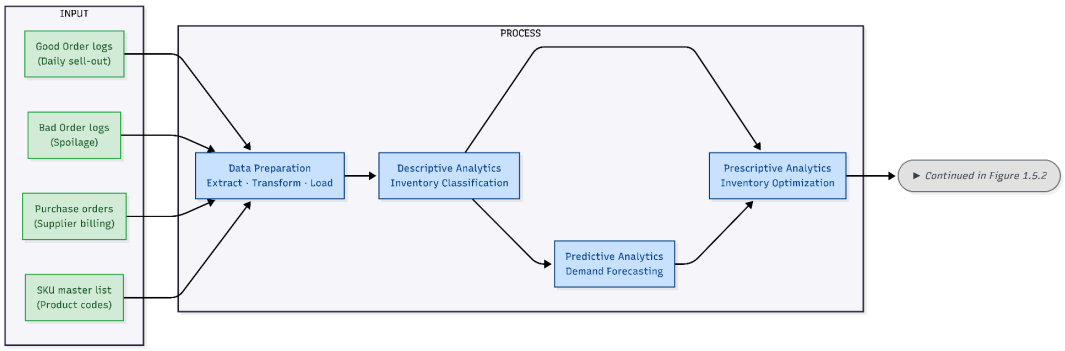


Figure 1.6.1. Input and process flow of the proposed Decision Support Dashboard, illustrating the data extraction, transformation, and loading stage followed by descriptive, predictive, and prescriptive analytics pipelines. (Continued in Figure 1.6.2.)

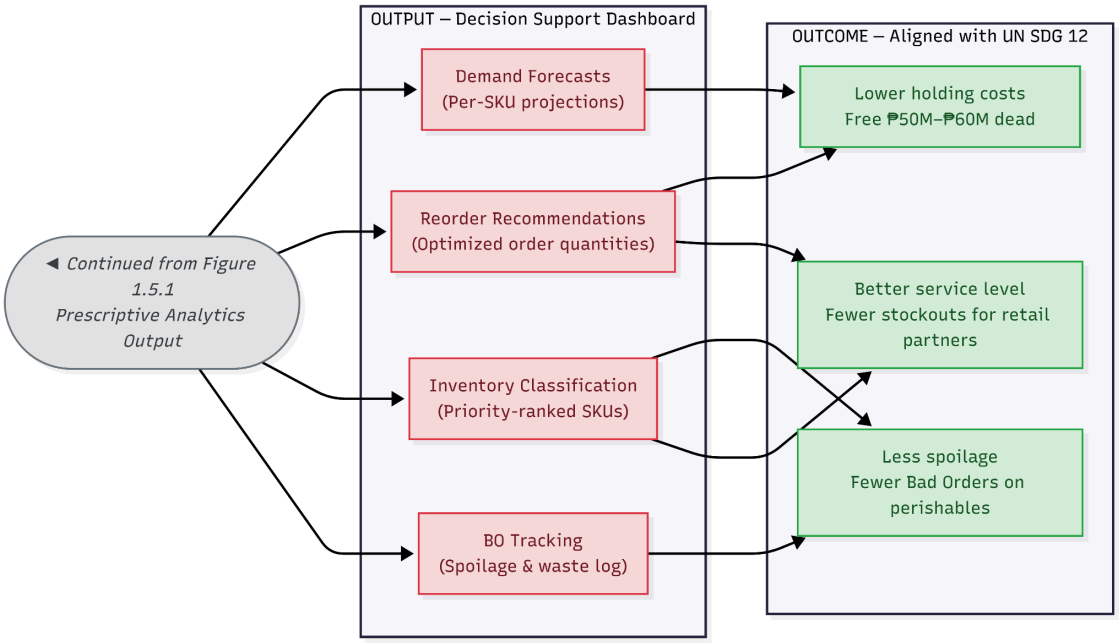


Figure 1.6.2. Output and outcome components of the proposed Decision Support Dashboard, showing the four dashboard modules and their corresponding operational outcomes aligned with UN Sustainable Development Goal 12: Responsible Consumption and Production. (Continued from Figure 1.6.1)

The proposed capstone project adopts an Input-Process-Output-Outcome (IOPO) model to illustrate how San San Marketing's historical data will transform into actionable inventory decisions, aligned with the United Nations SDG 12 on responsible consumption and production.

Input. The dashboard will draw from four data sources: Good Order logs containing stock on hand logs, Bad Order logs capturing waste incidents, purchase order files from principal suppliers, and a SKU master list serving as the central product registry.

Process. Data will pass through four sequential stages. Data preparation will clean and load transaction records into a structured database. Descriptive analytics then will classify all active SKUs into priority tiers based on their contribution to revenue and inventory movement. Predictive analytics will use the classifications to generate demand forecasts for high-priority products. Finally, prescriptive analytics will translate those forecasts into optimized replenishment decisions that balance holding costs and warehouse capacity.

Output. Results will be consolidated into a Decision Support Dashboard presenting four key outputs: per-SKU demand forecasts, optimized reorder recommendations, inventory classification results, and Bad Order tracking records. All recommendations require procurement staff approval before execution.

Outcome. The dashboard will target three improvements: reducing holding costs by releasing an estimated ₱50M–₱60M in trapped capital, minimizing product spoilage through demand-aligned replenishment, and improving service levels for retail partners by reducing stockout frequency.

1.7 Project Milestones / Timeline

Managing a complex two-semester project requires more than a simple calendar. The timeline presented operates as a functional guide for the team's Agile sprints. By mapping tasks systematically, the researchers can identify missing data dependencies early, long before any actual predictive modeling begins. To match the required academic deliverables, the entire schedule is split into two distinct halves.

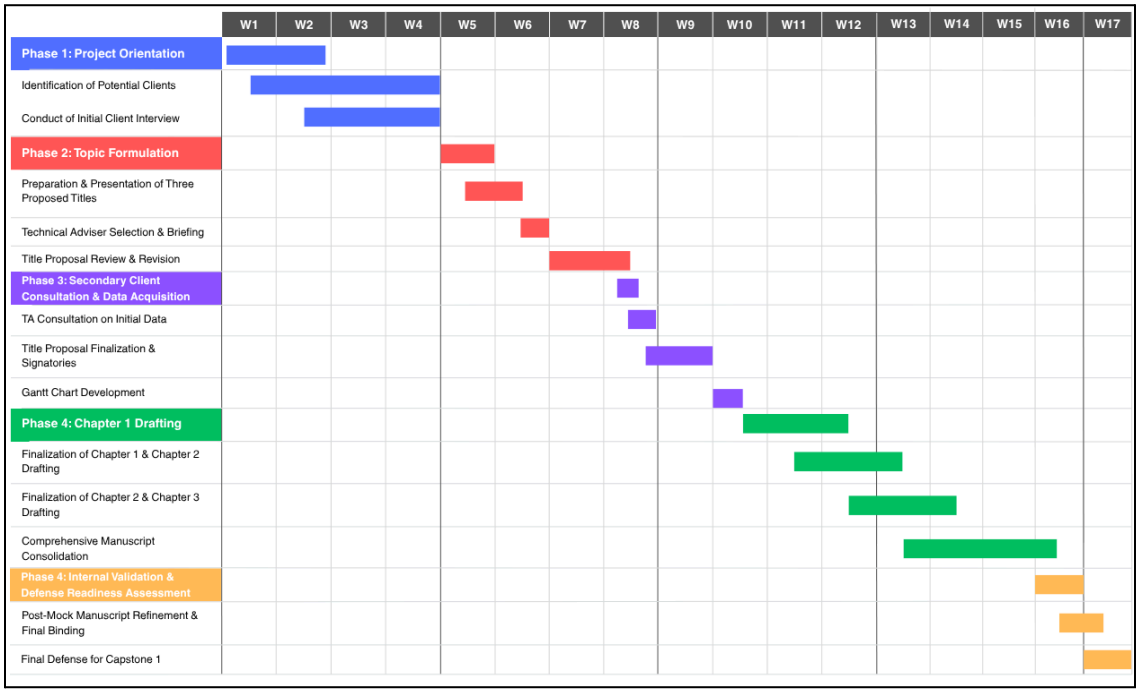


Figure 1.7.1: Capstone 1 Project Milestone

The succeeding phase of the research transitions from theoretical documentation to the active development of the proposed solution. The following semester will focus on the technical implementation of the prescriptive analytics models, the interaction of the DSD, and rigorous system evaluation.

Figure 1.6.1 outlines the anticipated development sprints required to build and test the DSD before the final Capstone 2 defense.

Chapter 2: Review of Related Literature, Studies, Systems and Technologies

This chapter provides a thorough review and critical analysis of literature, studies and systems pertinent to the development of Prescriptive Analytics for Multi-Objective Inventory Replenishment: A Decision Support Dashboard for San San Marketing. It summarizes and synthesizes earlier research, findings, and methods to help build a more complete understanding of the core concepts of the projects including the forecasting of seasonal demand and the development of mathematical models used to optimize inventory replenishment. By performing extensive research, this chapter lays the groundwork by providing examples of previous and familiar studies in the fields of supply chain management and data analytics, which assists the proponents in placing their study into context and identifying relevant new information that will support their moving from a fixed allocation of inventory to an automated, data-based decision-making process.

2.1 Review of Related Literature and Studies

This section of the work will provide an extensive review of the literature and research on developing Multi-Objective Inventory Replenishment Decision Support Dashboard through Prescriptive Analytics for San San Marketing.

Inventory Management in Supply and Distribution

Inventory Management is a pivotal process in supply and distribution management, an important asset that thoroughly handles a company's stock. The aim of Inventory management is to avoid disruptions to the company caused by out-of-stock situations. An effective inventory management system ensures there are

sufficient quantities of inventory to fulfill demand without overstocking. (Bautista & Young, 2021)

SSM currently relies on fixed allocation of inventory practices that lack systematic forecasting mechanisms. The stocks rely heavily on a manufacturer-driven replenishment model rule; the uniform policy, thus, does not account for demand variability across SKUs, leading to overstocking of slow-moving items and potential stockouts for high-turnover products.

The operational challenges underscore the need for a Decision Support Dashboard that can integrate historical sales data and apply predictive demand modeling. By shifting from static rules to dynamic, data-driven methodologies, the proposed dashboard aims to generate precise, SKU-level replenishment recommendations that optimize inventory flow and maximize capital efficiency.

Financial Impact of Overstocking and Understocking

In modern retail and supply chain management, inventory is not simply a collection of physical goods but a significant financial asset that reflects tied-up capital in a company's balance sheet. When businesses rely on outdated or manual methods for tracking and forecasting inventory, it can create serious financial risks. According to Wynn (2021), errors in manual inventory records can negatively affect an organization's overall financial condition. These inaccuracies may result in unaccounted losses, incorrect pricing decisions, and limited operational control. Since managers and procurement personnel depend on these records for decision-making, the use of unreliable data can ultimately lead to financial losses for the business.

When inventory systems fail to accurately represent current stock levels or forecast future demand, it can lead to serious imbalances within the business. These imbalances often appear in two critical forms: overstocking and understocking. Overstocking can strain a company financially by tying up capital in excess inventory and increasing holding costs, including storage expenses and the risk of product obsolescence. On the other hand, understocking or frequent stockouts can result in lost sales opportunities, reduced customer satisfaction, and disruptions in business operations.

Given the significant financial consequences of manual record errors and inventory imbalances, relying on guesswork or basic spreadsheets is no longer an effective strategy for businesses. Organizations need to shift from manual processes to more automated and data-driven approaches. By implementing a Decision Support Dashboard (DSD) that can automatically monitor inventory data and optimize reorder points using analytical models, businesses can address the underlying causes of poor inventory management. This approach helps reduce the financial losses associated with manual inaccuracies (Wynn, 2021) as well as issues related to overstocking and understocking, ultimately improving operational efficiency and supporting more informed decision-making.

Data Preprocessing in Forecasting Accuracy

In supply chain management, the reliability of a Decision Support Dashboard depends heavily on the quality of its historical data. In real-world situations, business records often contain errors, missing values, or are stored in unstructured formats, which can affect the accuracy of analysis. As Konda (2022) states, data quality is a

critical factor in any data-driven project, since the presence of redundant, noisy, or unreliable data can negatively impact the overall analytical process. Because of this, using raw, uncleaned data in forecasting models may lead to inaccurate predictions and ineffective inventory strategies.

To minimize these errors, a system should first perform data preprocessing before predicting future demand. Data preprocessing is an important initial step that transforms unstructured data into a usable format by addressing inconsistencies, errors, and missing values (Konda, 2022). In addition, Joshi and Patel (2020) highlight that the quality of the results depends on the quality of the input data. By properly cleaning and organizing the data beforehand, the system can improve the accuracy and efficiency of the forecasting model, allowing it to identify real market patterns instead of being affected by random errors.

For the proposed dashboard, proper data preprocessing is an essential step before performing predictive demand modeling. If statistical outliers, such as unusual spikes in sales or incomplete records, are not addressed, the model may misinterpret these irregularities as actual seasonal or cyclical patterns. This can lead to inaccurate forecasts, which may cause the system to suggest inappropriate replenishment quantities. By applying appropriate data cleaning and validation techniques, the system helps ensure that the results of the analysis are reliable and accurate. As a result, the generated recommendations can better support decision-making and help reduce the risks of overstocking and stockouts.

Analytical Frameworks for Inventory Replenishment in Distribution and Warehousing Operations

Inefficient replenishment decisions whether from overstocking, stockouts, or misallocated resources directly affect operational costs, warehouse capacity, and service levels. Several analytical frameworks have been developed to guide organizations in classifying inventory, understanding demand behavior, and evaluating the true cost of holding stock. These examines the key methods and frameworks relevant to inventory replenishment in distribution and warehousing contexts, including ABC/Pareto classification, sales velocity analysis, revenue-volume assessment, contribution margin evaluation, and SKU rationalization, providing a structured foundation for developing an effective and data-driven replenishment strategy.

ABC Analysis

Systematic inventory planning is a prerequisite for achieving operational balance, mitigating the risks of stockouts or overages, and curbing holding costs. Within this framework, ABC analysis serves as a fundamental methodology, segmenting inventory into distinct tiers derived from annual consumption rates or total carrying value (Silaen et al., 2024). The framework categorizes inventory into three distinct tiers based on annual consumption value. Often referred to as a Pareto analysis, this method serves as a critical instrument for organizations seeking to refine their inventory management strategies. It is rooted in the Pareto Principle (the 80/20 rule), which posits that approximately 80% of outcomes are derived from 20% of causes; in an inventory context, this highlights the necessity of prioritizing high-impact items (Jayakumaran et al., 2020).

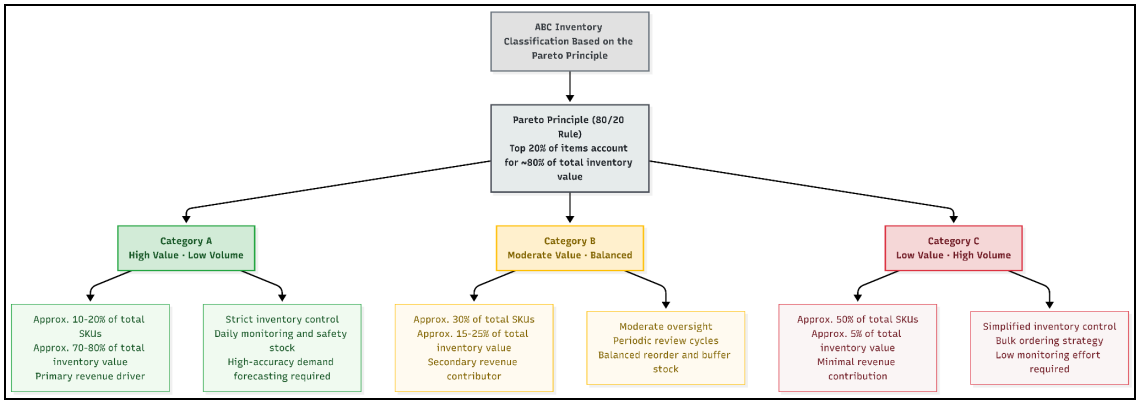


Figure 2.1 ABC Inventory Classification Based on the Pareto Principle

As what is shown in the figure, Category A items represent approximately 10–20% of SKUs but account for 70–80% of total inventory value; Category B items comprise around 30% of SKUs and contribute 15–25% of value; and Category C items make up the remaining 50% of SKUs while contributing only around 5% of total value. In an inventory context, this principle highlights the necessity of directing management attention and resources toward high-impact items, while applying simplified controls to lower-priority stock. While ABC Analysis provides the theoretical framework for product categorization, it will operationalize the logic through a multi-objective decision support dashboard. By integrating classifications into a centralized dashboard, it moves beyond static categorization to dynamic replenishment. However, while ABC analysis establishes a clear value-based hierarchy, it does not account for the frequency or predictability of demand, limitations addressed by the complementary frameworks discussed below.

Sales Velocity and Demand Patterns

Categorization alone does not capture the 'pulse' of the market; therefore, it is necessary to integrate Sales Velocity (how fast things sell) and Demand Patterns (the rhythm of sales) into the decision dashboard. By synthesizing the fiscal priority of ABC tiers with how fast a product sells and the predictability of consumption, the proposed dashboard can transition from simple categorization to precise, time-sensitive replenishment forecasting.

While sales velocity provides a quantitative measure of inventory depletion over a specific period, it remains a "lagging indicator" that reflects past performance. To achieve proactive replenishment, this velocity must be contextualized within established demand patterns (Hirche et al., 2021). Within the replenishment algorithm, velocity is not treated as a constant; rather, it is a weighted moving average that allows the system to adjust reorder points in real-time. This ensures that Category A items with accelerating velocity are replenished before a critical shortage occurs. Incorporating demand pattern analysis alongside velocity metrics therefore improves forecast accuracy and reduces the risk of both overstocking and stockout events.

Sales velocity in the proposed system is not treated as a standalone analytical module but is operationalized as an input variable within the ABC classification and SARIMA forecasting pipeline. The avg_monthly_qty field computed during the descriptive phase quantifies the rate of SKU depletion, which directly informs both the Pareto-based revenue classification and the demand signal fed into the SARIMA model. Rather than generating a separate velocity report, the DSD embeds velocity logic into its classification and

forecasting engines, ensuring that replenishment recommendations for Category A items reflect both their financial weight and their rate of market consumption.

Contribution Margin by SKU

While ABC analysis and demand pattern evaluation optimize the flow of goods through the supply chain, neither framework inherently accounts for the financial yield of individual items. Two SKUs may share identical sales velocity and ABC classification yet differ substantially in profitability. Contribution margin — defined as revenue minus variable costs at the SKU level — addresses this gap by providing a direct measure of each item's financial contribution to operational overhead and profit.

Integrating contribution margin into the replenishment framework enables more strategically aligned inventory decisions. High-velocity, high-margin SKUs merit prioritized storage positioning and faster replenishment cycles, whereas high-velocity, low-margin items may be consuming disproportionate operational resources relative to their financial return. This evaluation ensures that replenishment logic is driven not solely by volume or value classification, but also by profitability — a dimension that is particularly significant in distribution environments operating on thin margins.

The proposed system approximates the financial prioritization function of contribution margin analysis through its ABC classification engine, which ranks SKUs by their cumulative revenue contribution relative to total inventory valuation. While a full contribution margin computation, defined as revenue

minus variable costs at the SKU level, is outside the scope of the proposed system due to the absence of SKU-level cost data in SSM's existing records, the revenue-based ABC classification serves as a practical proxy that achieves the same strategic outcome: directing replenishment resources toward items with the greatest financial impact on operations.

The Role of Seasonality and Trend Analysis in Retail Demand

In any data-driven supply chain, the accuracy of future demand predictions relies entirely on how well the framework understands past purchasing behaviors. Accurately predicting this retail demand is fundamentally challenging due to the inherent presence of trends and seasonality within historical sales data (Hasan, 2024). Because purchasing behavior is recorded sequentially, demand forecasting must be treated as a time-series problem with recurring seasonal fluctuations (Abbasimehr & Nahari, 2020). Operationally, before our proposed analytical framework can project future sales, it must mathematically distinguish between a temporary spike (e.g., a holiday rush) and a permanent shift in baseline popularity. Without this separation, the forecasting algorithm will conflate statistical noise with genuine market movement.

Isolating these dynamics is critical for optimizing inventory, especially for perishable products (Namwad et al., 2024). Namwad et al. warn that relying on static averages rather than dynamic seasonal adjustments exposes businesses to severe financial risks. If the proposed Decision Support Dashboard fails to recognize a recurring seasonal decline, it will calculate replenishments based on outdated peak-season averages. This specific mathematical error results in ordering excessive

stock just as market demand drops, directly causing trapped working capital and product spoilage.

Therefore, recent research emphasizes explicitly modeling seasonal data to build specialized, highly accurate forecasting frameworks (Abbasimehr & Nahari, 2020; Hasan, 2024). In the methodology, this means seasonality acts as an active filter during the descriptive data preprocessing phase. Establishing this mathematically sound baseline ensures our predictive engine receives clean data, allowing the final dashboard to align its replenishment recommendations with actual market rhythms rather than algorithmic errors.

Demand Forecasting

While the analytical frameworks discussed above establish a structured basis for classifying inventory and prioritizing replenishment decisions, classification alone cannot determine when and how much to reorder. Effective replenishment requires not only an understanding of what items are valuable or fast-moving, but also a reliable estimate of future demand. This necessitates the integration of demand forecasting methods into the replenishment process. Over the past few decades, researchers have proposed and continuously refined a range of forecasting approaches to meet the varying requirements of different operational contexts. Traditional quantitative methods, including time series analysis, econometric models, and machine learning techniques, each offer distinct advantages depending on the nature of the demand signal being modeled. More recently, scholarly attention has shifted toward hybrid and combined forecasting models that integrate two or more methods, reflecting a growing recognition that no single approach is universally optimal across the diversity of SKU behaviors encountered in real-world distribution environments.

The selection of an appropriate forecasting method is a critical determinant of replenishment effectiveness. A foundational challenge in this regard is that no single predictive model performs optimally across all item types, demand conditions, and forecasting horizons — a limitation that has been consistently documented in the retail forecasting literature (E et al., 2022). The literature broadly organizes demand forecasting approaches into three categories: individual methods, hybrid models, and combination forecasts, each suited to different demand environments and operational requirements.

Individual Methods

Individual forecasting methods form the foundational layer of demand prediction and serve as the baseline against which more complex approaches are evaluated. Among these, time series methods are the most widely applied in retail and distribution contexts, valued for their computational efficiency, interpretability, and relatively low data requirements. Simple approaches such as naïve models and moving averages carry forward recent observed values to generate forecasts and are frequently used as benchmarks in comparative forecasting studies precisely because of their consistency across different demand environments (E et al., 2022). Exponential smoothing methods extend this logic by assigning exponentially declining weights to historical observations, allowing more recent demand signals to exert greater influence on the forecast — a property that improves responsiveness to gradual shifts in consumption patterns.

A more structured extension of this family is the Seasonal Autoregressive Integrated Moving Average model, commonly referred to as SARIMA. Unlike standard time series approaches that treat demand as a continuous signal, SARIMA explicitly models the seasonal structure of demand by incorporating seasonal differencing alongside autoregressive and moving average components. This makes it particularly well-suited to SKU-level demand that follows regular, calendar-driven cycles — such as fluctuations tied to promotional periods, holidays, or weather-dependent consumption behavior. Zhao (2024) demonstrated this applicability in an e-commerce retail context, finding that SARIMA effectively captured seasonal purchasing patterns and supported more accurate inventory replenishment decisions. Falatouri et al. (2022) further confirmed SARIMA's practical relevance in a retail supply chain setting, reporting that it outperformed Long Short-Term Memory neural networks for products exhibiting clear seasonal behavior, while remaining computationally accessible for operational deployment.

It is important to acknowledge, however, that SARIMA's strengths are bound by its assumptions. The model performs well when demand follows a stable, identifiable seasonal structure, but its accuracy diminishes under conditions of high volatility, nonlinear demand relationships, or irregular consumption patterns — all of which are common across the broader SKU assortments encountered in large-scale warehousing environments (E et al., 2022). These limitations do not render SARIMA unsuitable; rather, they define the conditions under which it excels and establish the rationale for complementing it with more adaptive approaches. As the literature reviewed in the following sections demonstrates, hybrid and combination methods were

developed precisely to address what individual models — including SARIMA — cannot capture alone. Recognizing this boundary is not a weakness in the framework but a sign of methodological maturity: the most effective replenishment systems do not rely on a single forecasting method universally, but rather match the method to the structural characteristics of the demand being modeled (E et al., 2022; Falatouri et al., 2022).

Hybrid Methods

In response to the limitations of individual models, hybrid forecasting approaches integrate two or more methods with complementary strengths to capture different structural features of demand simultaneously. Wang et al. (2022) developed a hybrid LSTM-ARMA model incorporating an error compensation mechanism, which allows the system to correct for systematic forecasting errors in real time. Their findings demonstrated that this hybrid approach improved forecast accuracy in a manufacturing demand context by addressing both the sequential dependencies captured by LSTM and the linear autocorrelation structure modeled by ARMA — two features that individual models address in isolation but not in combination.

The relevance of hybrid methods to warehousing and distribution replenishment lies in the structural complexity of SKU-level demand. In environments where individual items exhibit varying combinations of trend, seasonality, and irregular fluctuation, a single modeling approach is unlikely to perform consistently well across the full assortment. Hybrid models offer a more flexible architecture that can be tailored to the specific demand

characteristics of different item categories, particularly when integrated with the inventory classification frameworks discussed in the preceding section.

Combination Methods and Dynamic Model Selection

A third approach in the forecasting literature concerns combination methods, wherein the outputs of multiple individual models are averaged or weighted to produce a final prediction. The core argument for this approach is that combining models reduces the impact of any single model's errors, producing more robust forecasts across varying demand conditions. E et al. (2022) advanced this further by proposing a dynamic weighting framework that matches the forecasting method to the demand pattern of each SKU — classifying demand as smooth, intermittent, erratic, or lumpy based on demand volatility and frequency of zero-demand periods. Their findings confirmed that no single method dominates across all demand types, reinforcing the need for adaptive, SKU-level forecasting decisions.

This insight directly contextualizes the role of SARIMA within a broader replenishment system. Rather than positioning SARIMA as a universal solution, the literature supports its use as the method of choice specifically for SKUs exhibiting stable, seasonal demand patterns — where its structural modeling of periodicity outperforms both simpler time series methods and more complex machine learning alternatives (Falatouri et al., 2022; E et al., 2022). For SKUs with erratic or lumpy demand, combination or dynamic selection approaches are more appropriate. This delineation is not a limitation of SARIMA but a

clarification of where it delivers the most value within a diversified, demand-pattern-aware forecasting strategy.

Predictive Analytics for Demand Forecasting

Managing inventory in the Fast-Moving Consumer Goods (FMCG) sector requires precise demand forecasting. The operational conditions are unforgiving. Distributors must simultaneously handle seasonal shifts, strict perishable expiration dates, and hundreds of SKUs. Manual estimation simply cannot keep pace with this complexity. Because of this, time-series models applied to historical sales data have become critical for uncovering recurring market patterns that traditional rule-of-thumb buffers completely miss (Vijayalakshmi & Thiyagarajan, 2020). For seasonal FMCG environments specifically, the Seasonal AutoRegressive Integrated Moving Average (SARIMA) model is highly effective. In fact, replacing static inventory rules with SARIMA-based logic yielded holding cost reductions of up to 88% in recent retail studies (Aditia & Utama, 2025). This exact magnitude of improvement directly addresses the operational inefficiencies and trapped capital SSM currently faces under its blanket three-week ordering policy.

The suitability of SARIMA relative to competing forecasting approaches depends substantially on the seasonal structure of the underlying demand signal. To evaluate this, Falatouri, Darbanian, Brandtner, and Udokwu (2022) conducted the most directly applicable peer-reviewed comparison in recent literature, putting SARIMA against Long Short-Term Memory networks on daily sales data drawn from perishable retail SKUs. Their analysis found that

SARIMA outperformed LSTM specifically on products exhibiting pronounced seasonal behaviour, while LSTM showed an advantage only for products with stable, low-noise, non-seasonal demand a distinction that carries direct implications for SSM's product mix, which includes Del Monte food items whose sales are tied to Philippine national holidays, the Christmas season, and school enrollment periods. This finding is further supported by Nagalakshmi et al. (2025), who compared five forecasting models including ARIMA, LSTM, and hybrid ARIMA-LSTM across 35 product-level series and found that ARIMA remained competitive with LSTM on grocery staples with only approximately 3% MAPE difference a margin too small to justify the infrastructure and complexity costs that LSTM training demands, particularly for an organization like SSM that operates without dedicated machine-learning hardware.

SARIMA consistently comes out ahead of other models when looking at distribution environments. When Doğan (2025) tested SARIMA alongside Holt-Winters and Prophet on real retail data, SARIMA ended up beating the rest. The results make sense because Holt-Winters actually does fine on simple trends, but it completely breaks down once complex seasonal patterns appear. That specific weakness is exactly why it's not qualified for SSM. SSM’s five year demand history isn't simple; it is heavily driven by recurring annual peaks and complex purchasing cycles across the regions it covers. As for the Prophet model, it did put up competitive numbers, but getting it to actually work required way too much extra parameter tuning to be worth the operational hassle.

SARIMA does have its own limits when you look at it. A review of 72 retail forecasting studies done by Chowdhury, Paul, and Rozony (2025) shows that traditional models do well with stable demand, but they become less effective once they face highly volatile and promo-driven sales. The proposed system design actually keeps this limitation in mind. If ever SSM starts to track outside variables like promotional periods and holiday calendars, the current setup can just be extended into a SARIMAX model without needing a complete rebuild. At present though, SSM operates in a much simpler context since they rely on five years of historical data from stable FMCG products with predictable annual seasonality. They also don't have any dedicated machine-learning hardware. Under these exact conditions, the literature consistently points out SARIMA as the best and most defensible primary forecasting model to use.

Prescriptive Analytics in Decision Support Dashboard

While accurate demand forecasting is an important foundation, predictions alone are not enough to complete the inventory management process. Several existing studies emphasize the importance of using prescriptive optimization to balance different financial goals. A more comprehensive approach should connect predicted demand with supply chain reliability by using multi-objective optimization to determine the right order quantities and safety stock levels. It helps balance the costs of excess inventory and the risks of running out of stock (Tadayonrad & Ndiaye, 2023). Overall, businesses need to go beyond simply predicting demand and instead focus on optimizing responses to demand. By integrating prescriptive optimization with predictive analytics within a unified Decision Support

Dashboard, a highly reliable and actionable architecture can be established for dynamic inventory control.

Multi-Criteria ABC Classification in Inventory Optimization

Faltouri et al. (2022) indicate that hybrid forecasting methodologies do work effectively when an organization segments and groups its inventory. A good example of this can be seen when using a pure statistical method (such as SARIMA) compared to a more complicated deep learning method for predicting items that have different types of peak seasonal demand. Because of this, it is very important to segment all items into their class prior to making forecasts so that the correct items will be matched with the right type of forecasting method to provide the best possible outcome.

According to Zhao (2024), classifying SKUs depending upon demand patterns enhances the ability to manage inventory levels by determining better reorder points and better utilizing distribution warehouse resources. Research supports that when SKUs are segmented based upon prioritizing them, this allows for improved scheduling of production and improved forecasting accuracy, allowing for production schedules to be maintained. Research conducted on inventory systems within manufacturing firms by Junior et al. (2022) also provides evidence that despite some very sophisticated mathematical model developments, the use of ABC classifications along with time-series forecasting methods are the most widely used practice within manufacturing for managing inventories with a high volume.

Zietsman (2023) raises the issue of developing an integrated procedure to assist multi-objective decision making when two objectives (minimizing stock carrying costs and avoiding stock-outs) conflict with one another. The research finds that employing a sound product classification system when making purchasing decisions will reduce the inefficiencies associated with manually planning for stock. The research conducted by Sari et al. (2024) showed that by conducting ABC analysis using profitability and sales volume, companies could more easily determine which high-value items should be monitored closely and which should not be.

Further research on the differentiation between high-priority and low-priority items conducted by Teplicka et al., (2024) demonstrates that reclassifying high-priority items and reducing inventory levels directly releases trapped capital and improves a company's cash flow. Lastly,, Aditia and Utama (2025) indicate that enhancing forecast accuracy through the use of product classification and using classification systems will also aid in improving the overall efficiency of the organization (reducing interest expenses and waste operations) through the reduction of excess inventory.

Machine Learning Algorithms

Linear Programming for Inventory Management

While predictive models like SARIMA establish demand expectations, the translation of those forecasts into actionable replenishment decisions requires a prescriptive layer. Linear Programming (LP) has been widely validated as an effective tool to be used for that kind of purpose. Santos et al. (2024) illustrated that a multi-objective linear programming optimization model

can simultaneously manage up to five conflicting inventory constraints across various storage facilities, and Santos et al. (2024) revealed that algorithmic optimization significantly surpasses manual estimation in all tested scenarios. Their study also revealed a gap: the model depended on simulated demand data instead of authentic historical inputs, precisely the deficiency the study's SARIMA to Linear Programming pipeline aims to rectify.

Becerra, Mula, and Sanchis (2022) conducted a systematic review of quantitative inventory models. The authors found out how Linear Programming and Mixed Integer Programming are highly proficient and effective at minimizing holding costs while also reducing goods spoilage. Despite the effectiveness of the models, many existing frameworks fail to properly integrate demand uncertainty and forecasting into the optimization constraints. The identified research gap is exactly what the proposed capstone project aims to address directly. Feeding the SARIMA forecast into the Linear Programming constraints merges the process rather than treating forecasting and optimization as separate steps.

When choosing an LP formulation, Perez et al. (2021) compared deterministic LP, stochastic LP, and reinforcement learning for multi-echelon supply chain management. The researchers found out how deterministic LP still performed respectably even when placed against the more computationally intensive alternatives, especially for stable demand patterns with minimal fluctuations. The findings support the decision of the proposed capstone project to use deterministic LP with periodic batch processing. The approach is much

easier to implement and at the same time easier for the operations team of SSM to understand and follow through. In another study focused on physical warehouse optimization, Manoharan et al. (2022) combined an LP-based model together with an AHP-TOPSIS decision-making framework. The authors were able to achieve up to 100% warehouse utilization in their best scenario. Using storage capacity as a hard LP constraint is highly applicable to the actual warehouse management needs of SSM. Finally, a separate study by Perez et al. (2023) additionally cataloged the widespread use of LP inside Decision Support Systems across real supply chain operations. The survey confirmed how LP-based DSS tools are not just academic novelties but are in fact already standard industry practices. The work also provides diagnostic techniques applicable during the testing phase of the proposed capstone project.

CRISP-DM

The Cross-Industry Standard Process for Data Mining (CRISP-DM) is a widely adopted process framework that provides a structured and iterative approach to planning and executing data mining and analytics projects. Originally developed through a consortium of industry practitioners in the late 1990s, CRISP-DM organizes the data analytics lifecycle into six sequential but iterative phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment (Schröer et al., 2021)

For the proposed capstone project, CRISP-DM provides the overarching process structure that governs how SSM's business problem is

translated into analytical requirements, how raw inventory data is prepared and modeled, and how the resulting dashboard outputs are evaluated against real operational performance before deployment.

SEMMA

The SEMMA framework — an acronym for Sample, Explore, Modify, Model, and Assess — is a data analytics process model developed by the SAS Institute to guide practitioners through the sequential stages of transforming raw data into actionable analytical outputs. While SEMMA shares structural similarities with CRISP-DM, it places greater emphasis on the technical data preparation and modeling stages, making it particularly well-suited to projects where the quality and structure of historical data significantly influence the reliability of analytical outputs.

For the proposed DSD, SEMMA complements CRISP-DM by providing a more granular technical roadmap for the data preparation and model building phases, ensuring that the demand reconstruction process, ABC classification, SARIMA fitting, and LP optimization each follow a disciplined, assessable sequence before outputs are surfaced in the dashboard interface.

Decision Support Dashboard

According to Punia & Sing (2020), within the domain of supply chain management, firms across the value chain spanning manufacturing, distribution, and retail frequently encounter the dual challenges of inventory optimization and resource constraints. The primary dilemma involves

determining the optimal procurement or production volume for products characterized by short life-cycles and high demand volatility prior to a finite selling season. This challenge creates a precarious balance: an overestimation of demand leads to significant obsolescence costs and capital tied up in excess inventory, whereas an underestimation results in stockout penalties, including lost revenue and the erosion of consumer goodwill (Laping et al., 2023).

Framiñán et al. (2024) demonstrated that embedding an optimization model directly within an interactive dashboard deployed for a large FMCG company significantly outperforms simple allocation rules in matching inventory to customer demand. Their framework, built for Heineken Spain's brewery operations, used a model-based order promising engine embedded in a dashboard interface to maximize service levels under supply-demand mismatches. The results confirmed that a model-driven dashboard produces superior service level outcomes compared to the First-Come-First-Served policies typically used in manual procurement operations. For the proposed SSM project, this study validates the core architectural decision of embedding optimization directly into the dashboard layer rather than treating forecasting and optimization as separate offline processes.

Iurasova (2025) developed an interactive dashboard for retail chain assortment management that optimizes product allocation and purchasing decisions across a retail network using a consolidated profitability coefficient computed from daily sales and inventory data. The algorithm was implemented in a real apparel chain and produced an average profitability growth of 1.61%

without requiring substantial investment in equipment or staff retraining. The study also explicitly noted that automated dashboards cannot fully replace human oversight — the tool generates recommendations and draft purchase documents, but strategic decisions require manual review. This positions the dashboard as a decision-support layer rather than an autonomous tool, which directly parallels the design philosophy of the proposed SSM DSD, where all reorder recommendations require procurement staff approval before execution.

Pavon Arnaiz et al. (2023), writing in a Philippine context, reinforced that time-series forecasting integrated within an inventory management dashboard produces measurable improvements in stock visibility and demand prediction accuracy for local businesses. Critically, their work also emphasized that technology alone cannot replace the human expertise needed for supplier relationship management and responsive supply chain decisions, a framing that directly informs SSM's scope constraint against automating purchase order placement and positions the proposed DSD as a decision-support tool rather than an autonomous procurement interface.

Collectively, the cluster of literature confirms that an integrated dashboard — one that connects historical data processing, demand forecasting, and inventory optimization within a unified interface — is the appropriate architectural response to SSM's manual planning problem. The proposed DSD adopts this validated structure and adapts it to SSM's specific FMCG distribution context in Northern Luzon.

2.2 Review of Related Systems and Technologies

The development of the proposed Decision Support Dashboard (DSD) for San San Marketing draws upon a range of established technologies and software frameworks that have been widely adopted in data engineering, predictive analytics, and interactive system design. This section reviews the relevant literature and documented applications of each technology as they pertain to the dashboard's architecture and analytical objectives.

Python

Python as an Analytical Programming Language

Python has emerged as the dominant programming language for data science, machine learning, and analytical systems development due to its extensive ecosystem of open-source libraries, readability, and cross-platform compatibility. Sklearn, statsmodels, and SciPy collectively provide robust implementations of statistical and optimization algorithms that are directly applicable to demand forecasting and inventory optimization tasks. For the proposed system, Python serves as the core processing language across all stages of the analytics pipeline, from data extraction and cleaning to SARIMA model training and Linear Programming computation.

Pandas and NumPy for Data Manipulation

Tabular data is notoriously messy to process from scratch, which is where pandas comes in. McKinney (2010) introduced DataFrame structures specifically to make manipulating this kind of data practical. Paired with NumPy's high-performance numerical computing, pandas forms the foundation

for data preparation. Specifically, it ingests the company's transaction records, aligns the SKU data over time, and calculates the necessary descriptive statistics.

Beyond basic formatting, pandas also handles a critical operational step: the ABC classification. Based on the Pareto Principle, this routine uses vectorized operations to sort inventory into three tiers, identifying the 20 percent of SKUs that drive roughly 80 percent of the revenue (Category A). That specific output dictates exactly which items are passed forward into the forecasting models.

Statsmodels and SARIMA Implementation

The statsmodels Library provides a comprehensive suite of statistical modeling tools for Python, including robust implementations of AutoRegressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) models. Seabold and Perktold (2010) introduced statsmodels as a library designed to bridge the gap between econometric modeling and practical data analysis. The SARIMA implementation within statsmodels has been widely applied in demand forecasting studies across retail and distribution contexts. Its support for seasonal parameter specification, information criterion-based model selection, and rolling forecast generation makes it particularly suited for modeling the periodic demand patterns observed in Fast-Moving Consumer Goods (FMCG) inventory environments. For the proposed dashboard, statsmodels provides the primary framework for generating and validating SKU-level demand forecasts against historical observations.

SQL Alchemy

SQLAlchemy is a Python-based Object-Relational Mapping (ORM) toolkit that provides a high-level abstraction layer between Python application code and relational databases. In analytical pipeline architectures, SQLAlchemy is commonly used to programmatically write model outputs — such as forecast values, reorder points, and safety stock levels — to relational tables and to retrieve them for dashboard rendering. Its integration with pandas further streamlines the pipeline by enabling direct read and write operations between DataFrame objects and database tables through the read_sql() and to_sql() methods. For the proposed DSS, SQLAlchemy serves as the data access layer connecting the Python analytics pipeline to the underlying relational database.

PostgreSQL or MySQL

PostgreSQL and MySQL as Relational Database Systems

For persistent storage, a relational database is essential. Platforms like PostgreSQL and MySQL are industry standards for maintaining centralized repositories of analytical outputs, ensuring the dashboard can query pre-computed forecasts without rerunning models on the fly.

The structure of that database is equally important. A simple star schema works well for a single process, but San San Marketing's source data spans five totally different document types: Sales Registers, Inventory Sign-Offs, Sell Out reports, Purchase Orders, and Sales Returns. Forcing all of that into one fact table would create massive data gaps. Instead, the DSS uses

a Galaxy Schema, or Fact Constellation. This design allows the five different fact tables to link to shared, conformed dimensions like Product, Date, and Supplier. Because the product SKU code remains consistent across the entire system, it becomes possible to cross-reference an incoming purchase order directly against a SARIMA demand forecast.

ETL Pipeline Architecture

Extract, Transform, Load (ETL) pipelines form the data integration backbone of modern analytical systems, ingesting raw source data, applying transformation logic, and loading processed records into a target data store for downstream analysis. For the proposed DSS, the ETL pipeline extracts daily transaction records from San San Marketing's Excel-based GO and BO log files, applies data-cleaning and standardization transformations, and loads the processed dataset into the relational database. The pipeline is designed to run periodically to ensure the DSS dashboard reflects up-to-date inventory performance without requiring manual data intervention. This automated ingestion mechanism is a fundamental architectural component that ensures the system remains operationally relevant as new transaction data accumulates over time.

Visual Analytics and Interactive Interfaces

A dashboard is not merely a data visualization tool but the front end of a Prescriptive DSS. By integrating predictive (forecasting) and prescriptive (optimization) modeling, the system enables firms to move from "reactive" to "proactive" management. Visual analytics integrates interactive visualization with analytical

computation to support human reasoning and decision-making in complex, data-rich environments. Python-based visualization libraries such as Plotly and Dash have gained widespread adoption for constructing interactive analytical dashboards due to their support for dynamic charts, filtering controls, and web-based deployment. A dashboard is not merely a data visualization tool, but the front-end of a prescriptive DSS. By integrating predictive modeling outputs — specifically SARIMA-generated demand forecasts — with prescriptive optimization results — namely LP-derived reorder points and order quantities — the dashboard enables organizations to transition from reactive inventory management, in which decisions are made in response to stockouts or spoilage events, to proactive management, in which replenishment actions are taken in anticipation of forecasted demand.

2.3 Synthesis

The literature reviewed across inventory management, demand forecasting, prescriptive optimization, and decision support systems provides both the theoretical foundation and practical justification for the proposed Decision Support Dashboard for San San Marketing.

A consistent finding across the reviewed studies is that manual and assumption-based inventory practices are insufficient for organizations operating in dynamic, high-SKU distribution environments. A study by Wynn (2021) demonstrates that static inventory rules, such as SSM's uniform three-week safety stock policy, result in recurring inefficiencies. These include overstocking that ties up working capital and stockouts that diminish service levels for retail partners. The literature confirms that the shift from intuitive guesswork to data-driven, model-supported decision-making is not

merely an operational improvement but a strategic necessity for FMCG distributors managing hundreds of SKUs across multiple supplier relationships.

The reviewed frameworks for inventory classification consistently point to ABC analysis, grounded in the Pareto Principle, as the foundational segmentation method for directing computational and managerial resources toward high-impact SKUs (Silaen et al., 2024; Jayakumaran et al., 2020). The proposed dashboard adopts ABC analysis as its sole classification method, reflecting SSM's primary operational concern and the dashboard's deliberate focus on Category A items where the greatest concentration of trapped capital and spoilage risk is documented. This focused approach ensures that the dashboard's forecasting and optimization engines concentrate exclusively on the SKUs with the greatest financial and operational consequence, preventing computational and managerial resources from being misallocated toward inventory with minimal impact on SSM's overall performance.

Sales velocity, while reviewed as a standalone analytical concept in the literature, is not implemented as a separate module in the proposed dashboard. Instead, it is operationalized as an embedded input within the ABC classification and SARIMA forecasting pipeline. The ABC classification serves as a practical and defensible proxy, directing replenishment resources toward items with the greatest financial impact on operations and achieving the same strategic prioritization outcome.

The demand forecasting literature reviewed in this chapter supports the selection of SARIMA as the primary predictive model for SSM's Category A inventory. Across multiple comparative studies — including Falatouri et al. (2022), Doğan (2025), and Nagalakshmi et al. (2025) — SARIMA consistently demonstrated superior performance over LSTM, Holt-Winters, and Prophet specifically for products exhibiting

stable, recurring seasonal demand patterns, which characterizes SSM's product mix Del Monte food items and Unilever personal care products tied to Philippine national holidays and calendar-driven consumption cycles. Critically, the literature does not position SARIMA as a universal solution. E et al. (2022) and Chowdhury et al. (2025) confirm that no single model dominates across all demand types, and hybrid or combination approaches are more appropriate for erratic or lumpy demand profiles. The proposed system's scope constraint, limiting SARIMA application to Category A items with identifiable seasonal structure, directly reflects this nuance, ensuring that the forecasting method is matched to the structural characteristics of the demand being modeled rather than applied uniformly across the full SKU assortment.

The prescriptive layer of the proposed system is grounded in a well-documented tradition of Linear Programming applications within inventory and supply chain optimization. Santos et al. (2024), Becerra et al. (2022), and Perez et al. (2021) collectively confirm that deterministic LP formulations are effective at minimizing holding costs while enforcing operational constraints such as warehouse capacity and product perishability, and that this approach remains practically superior to more computationally intensive alternatives in stable-demand environments.

The literature establishes a coherent, evidence-based architecture for the proposed system: ABC classification as the segmentation foundation that concentrates analytical effort on high-priority SKUs, SARIMA as the demand signal generator matched specifically to the seasonal structure of SSM's Category A products, deterministic LP as the prescriptive optimization layer that translates forecasts into operationally constrained replenishment decisions, and an interactive dashboard as the human-facing interface through which all recommendations are reviewed and

approved before execution. Each component is independently validated in the literature, and their integration as a unified pipeline addresses a gap that none of the reviewed systems has fully resolved in an FMCG distribution context comparable to SSM's operating environment in Northern Luzon. The proposed DSD therefore represents not merely a collection of established methods but a purposefully integrated analytical system designed to transform SSM's procurement process from reactive manual estimation into a resilient, data-backed operation aligned with the operational realities of regional FMCG distribution in the Philippines.

Chapter 3: Methodology

This chapter presents the research methodology adopted in the design and development of the proposed Prescriptive Decision Support Dashboard for San San Marketing. It outlines the research design, data collection procedures, analytical models, and system development approach that collectively guide the study from raw data acquisition to the generation of actionable inventory recommendations.

3.1 Development Methodology

This section details the step-by-step approach that will be applied in designing, developing, and evaluating the Decision Support Dashboard (DSD) for San San Marketing. To ensure a structured and iterative development process, the project will adopt a combination of Agile, CRISP-DM, and SEMMA methodologies. Agile will be applied in the development of the dashboard as it supports continuous improvement and flexibility throughout the dashboard-building process. Meanwhile, CRISP-DM will serve as the main framework for the overall data mining cycle, from understanding the business problem up to model deployment. SEMMA will be incorporated within the process to handle the more detailed stages of data preparation and modeling, including sampling, exploring, modifying, modeling, and assessing the data.

DSD Development Methodology

Agile Methodology

The development of the proposed capstone project will adopt an iterative Agile process, enabling incremental development, continuous refinement, and adaptability to evolving project requirements. Rather than building the entire dashboard at once, the proponents will build each

dashboard component cumulatively in cycles. Each cycle includes a new set of features and added functions, along with continuous testing and prototyping to verify that the project is functioning correctly before proceeding to the next phase.

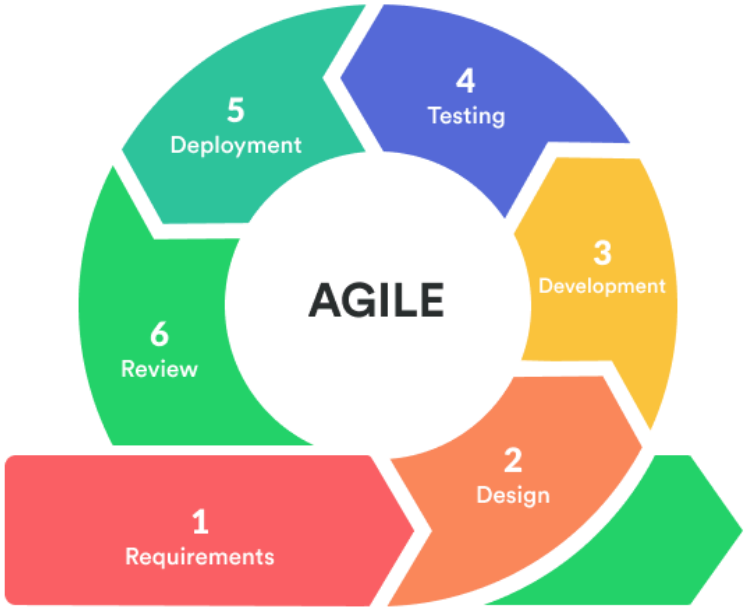


Figure 3.1 Agile Methodology

CRISP-DM

This proposed capstone project will utilize the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework. All components of this proposed capstone project will be planned using this method to ensure its objectives and goals are achieved from beginning to end.

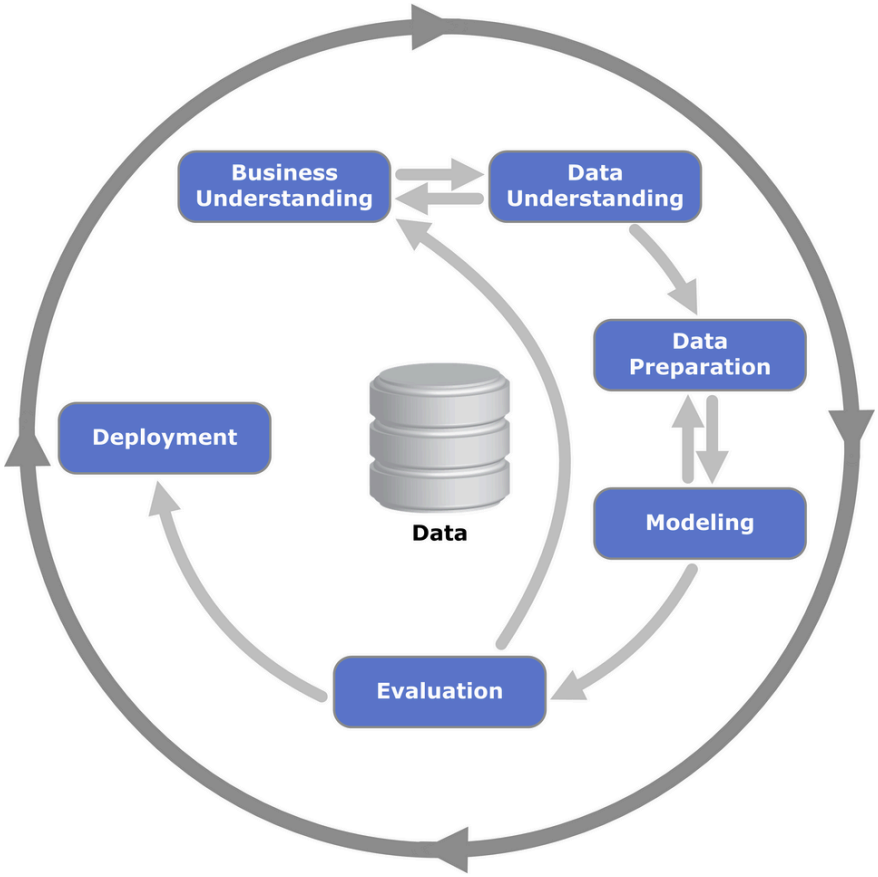


Figure 3.2 CRISP-DM Process

SEMMA

The SEMMA framework (Sample, Explore, Modify, Model, Assess) will be used to complete the proposed capstone project. All aspects of this project will be performed according to this framework so that data can be appropriately converted into actionable information and the final dashboard can achieve all of its objectives throughout all phases of the project.

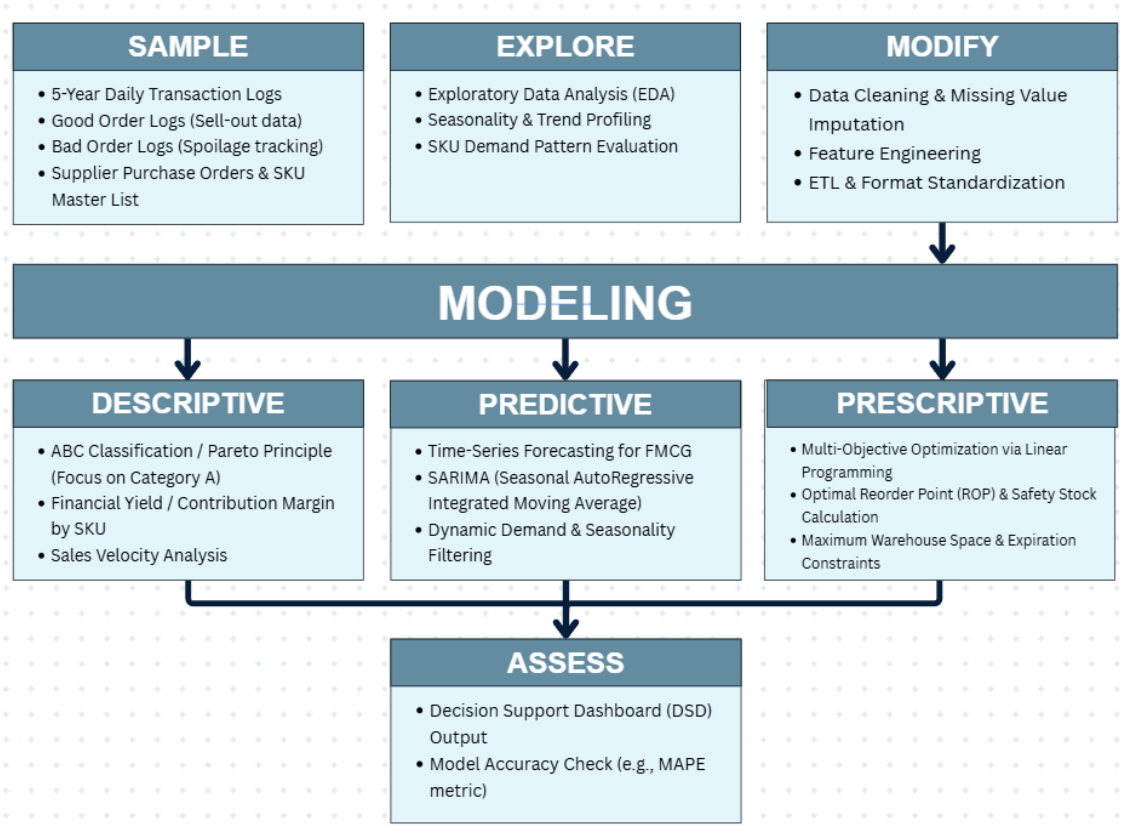


Figure 3.3 SEMMA Framework

3.2 Understanding the Data Gathering

The dashboard's data structure will use an ETL (Extract, Transform, Load) framework to connect various source systems. The dashboard will collect raw sales history and inventory logs, clean and standardize them, and store them in a centralized database. Having these processes in place establishes a reliable source of high-quality data for the descriptive charts, predictive demand forecasts, and prescriptive reorder models used by this pipeline.

Data Pipeline

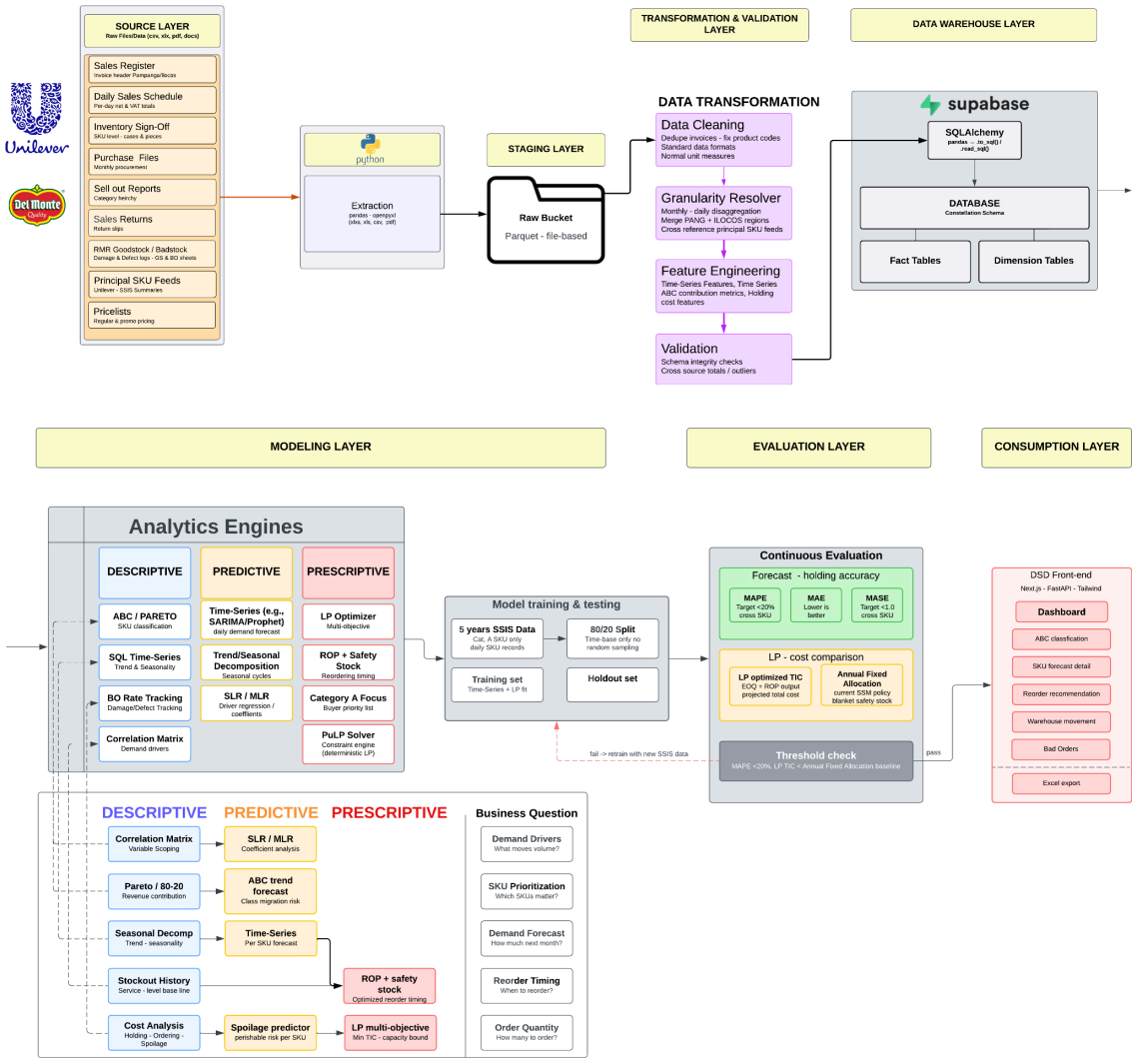


Figure 3.2 Decision Support Dashboard (DSD) Pipeline

3.2.1 Extract

The data pipeline starts first at the extraction phase where all source files like CSV, XLSX, and PDF files will be pulled from SSM and its suppliers (Unilever and Del Monte). The files will go through directly to a central raw bucket to make sure that the whole ingestion process stays completely standardized.

3.2.2 Transform

After extraction, the staged data is transferred to a Python environment for cleaning and standardization. Initial preprocessing focuses on resolving basic errors, such as removing duplicate invoices, correcting product codes, and adjusting unit measures to ensure consistency. Temporal restructuring is also performed; because the predictive models require continuous daily observations, aggregated monthly records are disaggregated into daily intervals. Feature engineering follows, generating advanced variables such as holding costs and ABC classifications. Finally, the dataset undergoes strict schema validation and outlier detection to ensure its reliability before proceeding to subsequent analytical stages.

3.2.3 Load and Data Warehousing

After the transformation stage, the cleaned dataset is loaded directly into the Data Warehouse layer. SQLAlchemy is used to securely write the structured data into a Supabase PostgreSQL database. The warehouse is organized using a Fact Constellation Schema, structuring historical records around core dimensions such as SKU, Date, and Supplier. This design substantially improves query efficiency and supports rapid access for downstream analytical models.

3.2.4 Visualization

The final phase brings all this warehoused data together to build an interactive dashboard. This interface takes all the descriptive, predictive, and

prescriptive numbers from the backend and turns them into real-time visual indicators. With this setup, SSM can actually evaluate how their inventory is performing and check their supplier allocations using high-quality, reliable data.

3.2.5 Tools or Technologies Used

Building the Decision Support Dashboard (DSD) requires a defined set of tools and technologies. Python serves as the primary development environment, with Pandas and NumPy handling data extraction, cleaning, and transformation. For storage, a PostgreSQL database hosted on Supabase is used, enabling the DSD to retrieve historical records accurately for analytical processes. The final output is an interactive web interface through which users can monitor key performance indicators and access real-time reorder recommendations, enabling San San Marketing to reduce trapped capital and avoid stock-outs.

3.3 Presentation of Constellation Schema



Figure 3.3.A SSM Constellation Schema



Figure 3.3.B SSM Constellation Schema Continuation

During the data modeling phase, San San Marketing's operational data is restructured into a Fact Constellation Schema. This architecture was selected because the organization's workflows encompass multiple distinct processes such as sales, purchasing, and inventory tracking that cannot be contained within a single fact table. By sharing common dimensions, the schema ensures analytical consistency and enables rapid Online Analytical Processing (OLAP) querying for the Decision Support Dashboard.

3.3.1 Fact Tables

The fact tables store the core quantifiable metrics and transactional records of the system.

Table 3.3.1 Fact Tables

Table	Description
Inventory_Fact	Records daily stock levels across different units of measure and computes the financial valuation of the warehouse inventory.
Sales_Fact	Captures outgoing customer transactions, recording net amounts, taxes, and applied discounts per invoice.
Purchase_Fact	Tracks incoming procurement from principal suppliers, logging purchase order details and billed amounts.
SellOut_Fact	Monitors the regional depletion of goods, grouping sold quantities by product hierarchies and areas.
SalesReturn_Fact	Logs transactions where products were returned by customers, documenting the refunded amounts and return slips.

ABC_Classification	Categorizes SKUs based on their historical revenue contribution, ensuring high-priority items are easily queried.

3.3.2 Dimension Tables

The dimension tables provide the descriptive, categorical, and temporal context required to filter and aggregate the fact data.

Table 3.3.2 Dimension Tables

Table	Description
Product_Dimension	Contains static details for all SKUs, including product codes, brand hierarchy, and unit measurements.
Date_Dimension	Provides a temporal hierarchy (days, months, quarters, and holidays) critical for time-series aggregation.
Supplier_Dimension	Stores vendor metadata, including contact information, Minimum Order Quantities (MOQ), and lead times.
Customer_Dimension	Maintains buyer profiles, categorizing clients by branch codes, towns, and regions.

Salesman_Dimension	Maps sales personnel to their assigned territories and operational codes.

3.4 Business Analytics Model and Testing

This section will describe a wide range of business analytics models under development to support the successful implementation of the DSD's strategic goals. Applications of these models include inventory classification, predictive forecasting, and prescriptive optimization. All of the analytic model output will undergo rigorous testing and validation, demonstrating that factual statements about San San Marketing's business operations can be generated using these advanced analytical models and supporting data-driven decisions that minimize trapped funds and spoilage and avoid stockouts.

3.4.1 Descriptive Analysis

The Decision Support Dashboard (DSD) will use historical sales logs and warehouse records to create full baseline metrics of how well inventory has performed at San San Marketing. The dashboard will incorporate interactive visual modules that allow managers to explore historical trends in product movement and identify instances of excessive spoilage or stock-outs. This descriptive layer will enable managers to identify specific categories of SKUs that have continually contributed to trapped capital, providing an overall, clear, data-based picture of past stock behavior and operational inefficiencies.

3.4.1.1 Data Overview

The SSM dataset contains historical transaction registers, inventory audit logs, and "Bad Order" (BO) reports. The SSM data set includes five (5) years historical records, spanning 2020 to 2025, of data for various sales and inventory metrics such as revenue contributions and SKU-level depletion for products as well as overall spoilage rates for all products throughout the Northern Luzon region with its retail partners. Prior to any analytical computation, all records undergo the ETL process described in Section 3.2. The cleaned dataset is loaded into the PostgreSQL database structured according to the Fact Constellation Schema presented in Section 3.3.

3.4.1.2 Inventory Classification Analysis

The proposed dashboard will apply ABC analysis grounded in the Pareto principle to classify all active SKUs into priority tiers based on their cumulative revenue contribution. This classification serves as the strategic filter determining which products are passed forward into the predictive and prescriptive modules.

ABC Tier Assignment

Table 3.4.1.2 ABC Tier Assignment

Category	Cumulative Revenue Threshold	Description
A	0% – 80%	Highest value; subject to full predictive and prescriptive processing
B	80% – 95%	Moderate value; descriptive monitoring only
C	95% – 100%	Lowest value; minimal monitoring

Only Category A SKUs are carried forward into the SARIMA forecasting engine and the LP optimization model. Category B and C items remain visible in the descriptive dashboard layer for monitoring purposes but do not receive forecast or replenishment recommendations.

3.4.1.3 Bad Order Monitoring

Bad Order (BO) data is extracted from SSM's RMR GS & BO reports, which log quantities and estimated value of items

removed from sellable inventory. For each active SKU, the system computes the monthly Bad Order Rate:

BO Rate (%) = (Total Bad Order Units ÷ Total Units Received) × 100

SKUs with consistently elevated BO rates are flagged in the dashboard for procurement review. Historical BO trends are visualized as time-series charts per SKU in the Bad Order monitoring module.

3.4.2 Predictive Analysis

The predictive layer generates per-SKU demand forecasts for Category A items, providing the quantitative basis for evaluating whether SSM's scheduled manufacturer allocations align with actual expected demand. Rather than accepting weekly delivery volumes at face value, the dashboard compares forecasted demand against allocated quantities to identify SKUs where the allocation is likely to exceed or fall short of market need. Forecasts are built on historical sales patterns derived from SSM's five-year transaction records, capturing seasonal cycles and demand trends at the SKU level.

3.4.2.1 Time-Series Decomposition

The historical daily logs go through time-series decomposition before hitting the main forecasting algorithms. This step breaks down the sales data for Category A SKUs into three specific pieces: the long-term trend, the predictable seasonal bumps from things like holidays, and the random market noise left over. Isolating those components is necessary. Without it, the forecasting engine wouldn't be able to tell if a sudden spike in demand is just a random error or an actual change in consumer behavior.

3.4.2.2 SARIMA (Seasonal Demand Forecasting)

SARIMA (Seasonal AutoRegressive Integrated Moving Average) is the primary forecasting model applied to Category A SKUs. SARIMA explicitly models the seasonal structure of demand by incorporating seasonal differencing alongside autoregressive and moving average components, making it well-suited to the calendar-driven demand cycles observed in SSM's FMCG product mix.

The implementation follows a structured training and evaluation workflow. During the training phase, the model utilizes five years of historical transaction records and employs a strict 80/20 time-based split specifically avoiding random sampling to preserve the sequential integrity of the time-series

data. The implementation of these predictive models follows a structured training and evaluation workflow mapped directly to the system's architecture. First, the data undergoes trend and seasonal decomposition to isolate underlying market patterns from recurring seasonal cycles. During the training phase, the model utilizes five years of historical transaction records and employs a strict 80/20 time-based split specifically avoiding random sampling to preserve the sequential integrity of the time-series data. The first 80% of the dataset is used for model training, while the remaining 20% serves as the holdout set. Finally, to ensure the model remains reliable for inventory decisions, it undergoes continuous evaluation against the holdout set. The system specifically targets a Mean Absolute Percentage Error (MAPE) of less than 20% across SKUs, establishing a strict threshold check before the forecast is allowed to pass into the prescriptive optimization engine.

3.4.3 Prescriptive Analysis

The DSD will utilize prescriptive analysis to translate forecasted demand into detailed, specific procurement strategies to support San San Marketing by making recommendations based on operational constraints and lead times for items that are commonly reordered and require high sales velocity. The DSD will also provide interactive, user-friendly displays for San San Marketing to help guide their purchasing decisions each day, using data-driven information generated by the dashboard. As part of the prescriptive element, these

recommendations will be the DSD's output, providing concrete steps to achieve optimal stock levels and maximize the overall effectiveness of inventory.

3.4.3.1 Safety Stock

Safety stock provides a demand buffer to protect against forecast errors and demand variability, ensuring that SSM maintains sufficient stock to meet customer demand with a 90% service level as established through stakeholder consultation. The safety stock for each Category A SKU is calculated as:

$$SS_i = Z \times \sigma_{di}$$

Where:

- **SS_i** = safety stock quantity for SKU i
- **Z** = 1.28, corresponding to the 90% service level target confirmed by SSM's procurement management
- **σ_{di}** = standard deviation of historical monthly demand for SKU i, computed from the reconstructed demand time series

The standard deviation of monthly demand is calculated as:

$$\sigma_{di} = \sqrt{[(1/n) \times \Sigma (D_t - \bar{D})^2]}$$

Where D_t = actual monthly demand for SKU i at month t, $\bar{D}$ = mean monthly demand over the historical period, and n = total number of months in the historical record.

3.4.3.2 Reorder Point

Since SSM operates within a supplier-defined annual allocation cycle, the Reorder Point is defined as the inventory threshold at which the procurement team should flag the current period's allocation for action:

$$ROP_i = \hat{Y}_{(t+1,i)} + SS_i$$

Where $\hat{Y}_{(t+1,i)}$ = SARIMA forecasted demand for SKU i in the following month and SS_i = safety stock quantity for SKU i. When SSM's current on-hand inventory for a given SKU falls at or below its ROP, the dashboard generates a replenishment flag for management staff review.

3.4.3.3 Multi-Objective Linear Programming (LP) Optimization

The LP optimization model determines the recommended acceptance quantity Q_i for each Category A SKU per planning cycle. The model simultaneously pursues three objectives: minimizing total inventory cost, maximizing product availability, and minimizing spoilage cost from Bad Orders.

Decision Variable

Q_i = recommended acceptance quantity for SKU i from the supplier's allocation in the current planning cycle (units)

Objective Functions

The three objectives are combined into a single weighted composite objective function:

$$\textbf{Minimize } Z = w_1 \times TC - w_2 \times SL + w_3 \times SC$$

Where w_1, w_2, and w_3 are non-negative weights assigned to each objective reflecting their relative priority as established through stakeholder consultation, and the service level term is subtracted because maximizing availability is equivalent to minimizing its negative value.Initial weights were set at w1 = 0.45 (holding cost reduction), w2 = 0.35 (service level), and w3 = 0.20 (spoilage cost), derived from SSM's stakeholder consultation, which identified holding cost reduction as the primary operational concern given the estimated PHP 50M-60M in trapped capital

The individual objective components are defined as:

$$\textbf{Total Inventory Cost: } TC = \Sigma (h_i \times Q_i)$$

Where h_i = holding cost per unit per month for SKU i.

$$\textbf{Service Level (Availability): } SL = \Sigma (Q_i / \hat{Y}_{(t+1,i)})$$

Where $\hat{Y}_{(t+1,i)}$ is the SARIMA forecasted demand for SKU i in the coming month, measuring the proportion of forecasted demand covered by the recommended acceptance quantity.

Spoilage Cost: $SC = \Sigma (BO_rate_i \times Q_i \times c_i)$

Where BO_rate_i = historical Bad Order rate for SKU i and c_i = unit cost of SKU i derived from the Last Purchase Price field in SSM's Stock on Hand records.

Constraints

1. Warehouse Capacity Constraint: $\Sigma (Q_i \times V_i) \leq W$

Where V_i = storage volume per unit of SKU i and W = SSM's maximum available warehouse capacity. This ensures total accepted stock does not exceed physical storage limits.

2. Supplier Allocation Ceiling Constraint: $Q_i \leq A_i$

Where A_i = supplier's allocation for SKU i in the current planning cycle as defined in the annual 52-week schedule. This ensures recommendations do not exceed what the supplier has committed to deliver.

3. Demand Coverage Constraint: $Q_i \geq ROP_i$

This ensures the recommended acceptance quantity for each SKU is sufficient to cover the reorder point calculated in Section 3.4.4.2, guaranteeing that safety stock and forecasted demand are met before warehouse capacity optimization is applied.

4. Non-Negativity Constraint: $Q_i \geq 0$ for all i

Allocation Variance Flag

In addition to the LP optimization output, the prescriptive module computes an Allocation Variance flag for each Category A SKU:

$$AV_i = A_i - \hat{Y}_{(t+1,i)}$$

The LP model is solved using Python's PuLP library applying the simplex method. For each planning cycle, the solver produces an optimal Q_i value for every Category A SKU, which is written to the database and surfaced in the dashboard's prescriptive recommendation interface alongside the ROP, safety stock level, and Allocation Variance flag.

3.4.4 Model Testing and Training

The model training and evaluation will utilize machine learning approaches through an interactive process. This systematic process will confirm the mathematical reliability and accuracy of the analytical models that will be used for demand forecasting and inventory optimization. By validating historical data sets with these analytical models, the proposed capstone project will provide San San Marketing with the final recommendations based on data-driven findings.

3.4.4.1 Model Training

The Model Training phase will consist of an array of analytical algorithms and modeling techniques including ABC/FSN classification and SARIMA time series. In this phase of model training, the proponents will be working on SSM's five years of historical data which will be divided into 80% for training and 20% for testing, thus providing a thorough and unbiased assessment of the models being developed. The training subset will be used to calibrate the models to achieve the highest level of mathematical accuracy through hyper parameter optimization. This phase will involve selecting appropriate features, preparing the data for use in the modeling pipeline, and confirming that the models selected can effectively identify patterns associated with Demand seasonality; inventory movement, and physical waste of the product.

3.4.4.2 Model Evaluation

Once trained, the models will undergo evaluation to test their accuracy and predictive capabilities on unseen data. Several metrics will be used to assess the overall performance of the models:

3.4.4.2.1 Evaluation Metrics for Inventory Classification

This section will analyze how well the ABC Classification Model used in segmenting the SKU assortment will work. The different metrics below will assist in determining whether the

model has successfully identified high-impact items and allows the proponents to allocate analytical and computational resources to those items that are most important from a financial and operational perspective.

3.4.4.2.2 Pareto Principle Validation (80/20 Rule)

The evaluation of the Pareto Principle will determine if the top 20% of items sold generate around 80% of San San Marketing's revenues. The classification model is valid if the top quintile of goods generates cumulative revenues that are between 75% and 85%, as real-world distribution can be slightly different from the expected 80%/20% ratio. If the results fall outside of this range, the classification boundaries may need to be modified in order to better reflect the company's revenue base through the use of segmentation logic.

3.4.4.2.3 Cumulative Revenue Coverage Ratio

The Cumulative Revenue Coverage Ratio (CRCR) measures the proportion of total revenue attributable to each classification tier, and is calculated as follows:

CRCR = (Σ Revenue Contribution per SKU / Total Revenue) × 100%

Where SKUs will be ordered from highest individual revenue contribution to lowest; then when summed, the SKUs

will be analyzed against the following tier benchmarks: A Tier products must account for ≥80% of total revenue; B Tier products must account for between 10% of total revenue; and C Tier products must not exceed 5% of total revenue. These benchmarks will be monitored on a monthly basis, and if there is a classification drift of >±5 percentage points in any tier, then the boundaries of the tiers will be reevaluated in order to preserve classification integrity over time.

3.4.4.2.4 Inventory Turnover Ratio (ITR)

The Inventory Turnover Ratio (ITR) will help track the efficiency of San San Marketing in cycling through its inventory over a specific period. The ITR also indicates whether the ABC classification model has correctly identified fast and slow moving SKUs. The ITR is calculated as:

$$ITR = \frac{Cost\ of\ Goods\ Sold\ (COGS)}{Average\ Inventory\ Value}$$

Where Average Inventory Value is derived as:

$$Average\ Inventory\ Value = \frac{Beginning\ Inventory + Ending\ Inventory}{2}$$

For category A SKUs, a higher ITR is expected, which will confirm the model's identification of items that are fast-moving and of high value. Conversely, a lower ITR among category C SKUs will confirm that the model

has successfully identified items that are slow-moving and potentially obsolete.

$$DIO = \frac{365}{ITR}$$

The Days Inventory Outstanding (DIO) ratio will also be calculated to express inventory efficiency using time, which will complement the ITR number. A high DIO number associated with Category C items is an indicator that the items are likely dead stock, i.e., inventory that is no longer generating enough turnover versus its carrying cost. This is significant because San San Marketing is estimated to have between ₱50M and ₱60M in trapped capital; therefore, this measurement will provide San San Marketing with a direct, data-driven way to determine specifically which SKUs contribute to this figure the most and should be prioritized for liquidation or replenishment strategy alteration.

3.4.4.2.2 Evaluation Metrics for Time Series Forecasting

In order to assess the accuracy of forecasts generated, various time series error metrics will be utilized. Such metrics will show how well the expected forecasts of future customer demand align to the actual observed demand for customers over time. By using multiple metrics, the proponents will be able to measure both the total amount and the relative size of forecast

errors for all high-volume SKUs represented within the model in a comprehensive manner.

3.4.4.2.2.1 Mean Absolute Error (MAE)

The Mean Absolute Error determines the overall size of the errors involved with a given set of forecasts. This value does not take into account whether the errors are positive or negative. The Mean Absolute Error is a measurement of how far, on average, the demand that the model predicts is from the actual demand. A lower Mean Absolute Error indicates that the model has provided greater accuracy and reliability for making decisions regarding inventories.

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

Where:

y_t = actual value

$\hat{y}_t$ = forecasted value

n = number of observations

3.4.4.2.2.2 Mean Absolute Percentage Error (MAPE)

Forecasting accuracy can be expressed in percentage terms by using the Mean Absolute Percentage Error, allowing for much easier understanding and comparison with SKUs having

different levels of inventory and price. The MAPE provides a measure of how much actual values differ from forecasted values on average. This gives users a way to evaluate model performance on their SKUs in a manner that is consistent across whatever the size or nature of those SKUs are. It is important to note that MAPE may not be reliable in measuring forecasting accuracy for any SKU whose actual inventory quantity is very near to zero. When this is the case MAE and RMSE will be used as the key measures of performance for these SKUs.

$$MAPE = \frac{100}{n} \sum_{t=1}^{n} \left| \frac{y_t - \hat{y}_t}{y_t} \right|$$

3.4.4.2.2.3 Root Mean Squared Error (RMSE)

The RMSE squares errors before taking an average, thus providing more weight to larger errors, it is more likely to register larger deviations than traditional average as well as other error measures such as MSE. For example, if the SARIMA model were to occasionally produce larger than expected forecast errors, using RMSE will help to identify this potential occurrence leading to expensive stockout and excessive stockpiling situations for San San Marketing in the future. The

smaller the RMSE the closer the predictions from the model are to the actual observed values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

3.4.4.2.2.4 Mean Absolute Scaled Error (MASE)

The Mean Absolute Scaled Error provides a way to compare the performance of your SARIMA model to that of some simple naive base line forecast, e.g., using the previous period's inventory level or reorder point value. It enables an objective means of comparing performance across SKUs and across all time series, independent from their respective individual scales or units of measure.

$$MASE = \frac{\frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|}{\frac{1}{n-m}\sum_{t=m+1}^{n}|y_t - y_{t-m}|}$$

If a given MASE value is below 1.0, this indicates the performance of the SARIMA model in forecasting is better than that of a naïve method by showing that the model is capable of producing more accurate forecasts than using just the last observed value as the forecast.

3.5 Business Analytics Tools, Techniques, and Specific Applications

This section explains the technical framework and analytical models that will be used in developing the Decision Support Dashboard (DSD). To clearly explain how raw inventory data is transformed into actionable business strategies, this section is structured according to the core stages of analytics. It presents the software tools and analytical techniques used in data preparation, historical data visualization, demand forecasting, and inventory optimization. Furthermore, it examines how these analytical processes are implemented within the DSD, illustrating how backend analytics are translated into functional, user-facing features.

3.5.1 Data Preparation & Integration

Tools to be used:

- Python, Supabase

Techniques to apply:

- Perform Extract, Transform, and Load (ETL) to clean the raw transaction log data from Del Monte and Unilever, standardize it, and make it usable across entity boundaries.
- Use Python scripts to clean the historical logs and correct any discrepancies in the raw bucket staging area.
- Integrate Refined Data into Supabase to establish a centralized relational database that supports high-speed querying for the dashboard interface.

3.5.2 Descriptive Analytics

Tools to be used:

- Python (Pandas/NumPy), SQL, Supabase.

Techniques to apply:

- Perform a comprehensive analysis using ABC and Pareto methods to identify those particular product lines responsible for the highest levels of inventory volume and total amount of revenue.
- Conduct Demand Profiling to define product category groupings based on demand behaviour to develop more specialised strategies for product inventory management.
- Execute an Analysis of Historical Trends and Seasonality to analyze five years' worth of historical sales data to identify and isolate ongoing seasonal patterns.
- Utilize a Time-Series Trend Analysis approach to visually represent and track "Bad Orders" (B/O) by SKU over time to quantify actual product waste.

3.5.3 Predictive Analytics

Tools to be used:

- Python (Statsmodels), Supabase

Techniques to apply:

- Use the Regression Analysis method and perform both Multi-Linear Regression and Simple Linear Regression to determine what drives demand at each Product Level.

- Forecast Future Demand for each Category A SKU using Forecasting Techniques, such as Seasonal Autoregressive Integrated Moving Average (SARIMA).
- Measure Accuracy by using MAPE, RMSE, and MAE to compare predicted values against actual sales data.

3.5.4 Prescriptive Analytics

Tools to be used:

- Python (PuLP Solver), Supabase.

Techniques to apply:

- Use Linear Programming (LP) for an Optimized Model, determine the quantity of each SKU to reorder at SSM to satisfy demand.
- Calculate Reorder Points (ROP) and Safety Stock to determine when SSM needs to place a replenishment order for each SKU.
- Implement Multi-Objective LP Optimization to minimize the total cost per order of inventory while meeting the operational and storage limitations.

3.5.5 Specific Applications within the DSD

The Decision Support Dashboard exposes analytical outputs through five modules:

- The Inventory KPIs module displays current stock valuation, active SKU count, aggregate Bad Order rate, and dead stock value.

- The ABC Classification module shows cumulative revenue contribution per SKU tier and flags SKUs approaching tier boundaries.
- The SKU Forecast module presents SARIMA demand projections for Category A SKUs across a 4-week horizon with MAPE and RMSE indicators.
- The Reorder Recommendations module displays LP-optimized allocation acceptance quantities per SKU against the scheduled manufacturer allocation.
- The Bad Order Tracking module shows historical spoilage rates per SKU as a time series, enabling correlation with allocation periods.

3.6 Risk Assessment and Mitigation

In this section, the proponents present the comprehensive risk assessment conducted to anticipate potential challenges and uncertainties inherent in the proposed Decision Support Dashboard (DSD) for inventory management. The proactive measures outlined aim to mitigate the identified risks and ensure the DSS's reliability, accuracy, and operational effectiveness.

3.6.1 Risk Impact

Table 3.6.1 Risk Impact Rating

Impact Rating	Definition
High	Significantly affects system performance, decision accuracy, and business operations, requiring immediate and robust mitigation strategies.
Medium	Moderately affects system functionality and analytics output, requiring proactive mitigation measures.
Low	Minor effect on system performance, manageable through routine monitoring and standard controls.

Table 3.6.1 shows the Risk Impact Rating framework used to asses and classify possible vulnerabilities in the Decision Support Dashboard (DSD). For the system to remain reliable, it is important to identify and measure potential threats in a structured manner. A “High” impact rating is given to critical risks that can significantly affect system performance and decision-making accuracy, which means they need immediate and strong solutions. On the other hand, “Medium” and “Low” impact ratings refer to risks that cause moderate or minor issues, usually affecting routine functions and can be handled through regular monitoring and preventive measures. By grouping risks into these three categories, the framework helps ensure that major problems, such as errors in data integration or inaccuracies in predictive models, are addressed early before they can affect procurement operations.

3.6.2 Risk Likelihood

Table 3.6.2 Risk Likelihood

Impact Rating	Definition
High	Highly probable to occur during system operation, requiring continuous monitoring and preventive measures.
Medium	May occur under certain conditions, requiring periodic monitoring and mitigation strategies.
Low	Unlikely to occur but still possible, requiring minimal monitoring.

Table 3.6.2 presents the Risk Likelihood framework, which is used to measure how likely a specific vulnerability is to occur during the operation of the Decision Support Dashboard (DSD). A “High” likelihood rating is given to risks that are very likely to happen in normal conditions and require constant monitoring and strict preventive measures. Meanwhile, “Medium” and “Low” likelihood ratings refer to risks that may only happen under certain conditions or are rare, meaning they only need occasional or minimal monitoring. By evaluating the likelihood of risks, this approach helps the system become more proactive and efficient, ensuring that more attention and resources are given to the most frequent and critical risks

3.6.3 Risk Assessment

Table 3.6.3 Risk Assessment

Risk ID	Risk	Description	Impact	Likelihood	Mitigation
R01	**Data**	Historical	**High**	**High**	Implement

	Quality Issues	inventory and sales data may contain missing values, inconsistencies, or anomalies (e.g., sudden spikes), which can negatively affect forecasting accuracy.			automated data cleaning and preprocessing (handling missing values, outliers, and inconsistencies). Perform regular data validation and anomaly detection. Establish data governance policies to ensure consistent data entry and maintenance .

R02	**Data Integration & Preprocessing Failure**	The DSD fails to accurately ingest or clean raw inventory data from backend databases, feeding poor quality data into the predictice engine	**High**	**High**	Implement automated data cleaning and validation protocols before data enters the DSD. Apply Seasonal Adjustment Factors to normalize sudden demand spikes. Establish strict data governance for manual data entr
R03	**Predictive Engine Inaccuracy**	The DSD's underlying forecasting algorithms fail to capture	**High**	**Medium**	Use ABC Analysis to assign the most robust forecasting models to

		sudden market shifts or seasonal trends, generating flawed reorder recommendations.			high-priority Category A items. Continuously monitor the DSD's predictive accuracy against actual sales. Perform periodic algorithmic retraining and tuning.
R04	**Prescriptive Model Misconfiguration**	The optimization formulas within the DSD calculate reorder points that ignore physical warehouse limits or	**High**	**Medium**	Hardcode operational constraints (e.g., max warehouse capacity, strict shelf-life limits) directly into the DSD backend.

		product perishability rules.			Conduct regular batch testing using historical simulation data. Validate DSD outputs against real-world procurement scenarios.
R05	**Dashboard Misinterpretation & Low User Adoption**	Procurement staff find the DSD interface too complex or ignore the automated alerts, reverting to manual guesswork and causing inventory imbalance	**Medium**	**High**	Design an intuitive and user-friendly dashboard. Provide comprehensive training on how to interpret the DSD's Risk Impact Ratings. Frame the DSD as a collaborative tool to support, not

					replace, human decision-making.
R06	**DSD System Latency & Downtime**	The dashboard experiences server crashes or slow loading speeds during critical purchasing hours, delaying urgent replenishment orders.	**High**	**Low**	Deploy the DSD on a reliable, scalable cloud infrastructure. Implement robust API connectivity and error-handling mechanisms. Utilize data caching for high-velocity queries to ensure the dashboard loads quickly.

Table 3.6.3 presents the overall Risk Assessment for the proposed Decision Support Dashboard (DSD), showing the possible technical and operational risks that may affect the system. The matrix identifies “Data Integration and Preprocessing Failure” as the most critical risk because it has both high impact and high likelihood.

This is because inaccurate or unprocessed data can affect the forecasting results and reduce the accuracy of the system. To address this, automated data cleaning and validation processes are included in the system. In addition, risks such as “Predictive Engine Inaccuracy” and “Prescriptive Model Misconfiguration” are also considered high-impact since they may lead to unrealistic or incorrect procurement recommendations. To minimize these risks, the system will undergo continuous model evaluation, regular retraining, and the application of warehouse constraints within the system. Lastly, the assessment recognizes that the effectiveness of the dashboard also depends on the users, which is why user-friendly interface design and proper training are emphasized to reduce dashboard misinterpretation and encourage user adoption.

3.6.4 Risk Matrix

To combine the results of the impact and likelihood assessments, the proponents developed a Risk Matrix, as shown in Table 3.6.4. This matrix serves as a visual tool for prioritizing the identified risks in the Decision Support Dashboard (DSD). By comparing the severity of each risk with its likelihood of occurring, the matrix helps identify which parts of the system such as data integration, algorithm accuracy, and user adoption need more attention and stronger safeguards. The matrix also guides the development team in properly allocating time and resources, making sure that the most critical risks are addressed before the full implementation of the system.

Table 3.6.4 Risk Matrix

Likelihood / Impact	**Low**	**Medium**	**High**

Low			R06
Medium			R03, R04
High		R05	R01, R02

As illustrated in Table 3.6.4, the distribution of risks emphasizes the critical importance of data integrity and algorithmic precision within the DSD framework. Risks R01 (Data Quality Issues) and R02 (Data Integration & Preprocessing Failure) fall into the highest priority zone (high likelihood and high impact), confirming that automated data sanitization is the most urgent technical requirement for the project. Furthermore, mathematical threats such as R03 (Predictive Engine Inaccuracy) and R04 (Prescriptive Model Misconfiguration) are positioned in the high-impact column, dictating that rigorous model testing is necessary to prevent the system from generating faulty replenishment recommendations. Finally, while R06 (DSD System Latency & Downtime) carries a high financial threat, its placement in the low-likelihood row reflects the reliability of the system's planned cloud infrastructure. Ultimately, this matrix ensures that the project's mitigation strategies are strictly aligned with the operational realities of the business.

3.7 Mockups

This section presents the conceptual design of the proposed capstone project along with practical applications, using mock-ups for the Decision Support Dashboard (DSD). The DSD mock-ups show how the DSD will be set up, as well as describing several different ways the DSD can help to meet the inventory replenishment optimization objectives.

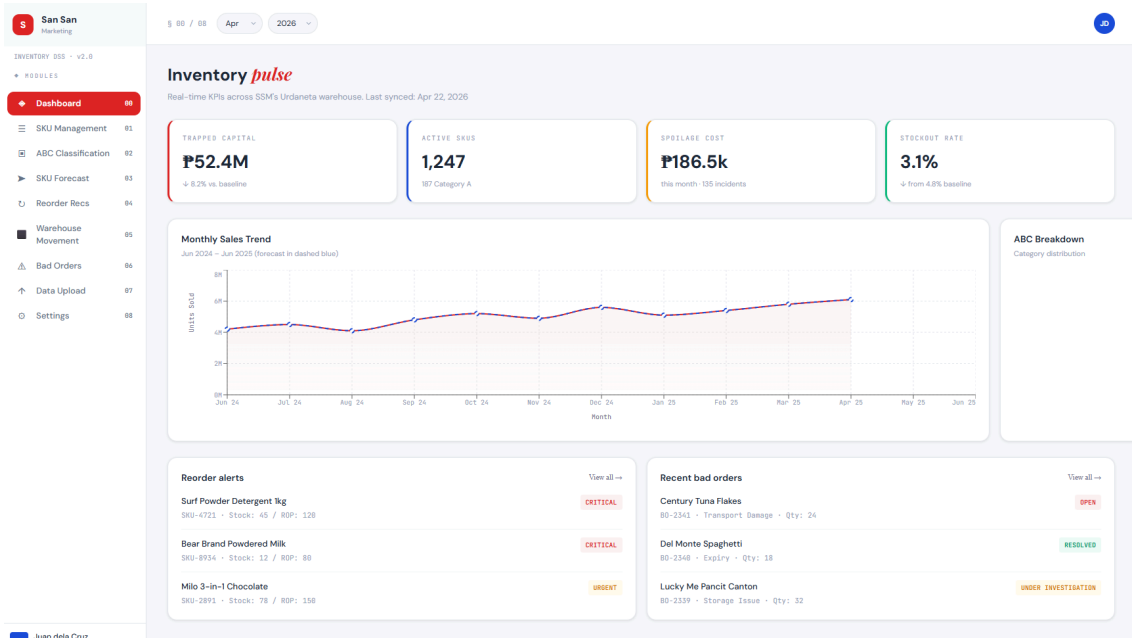
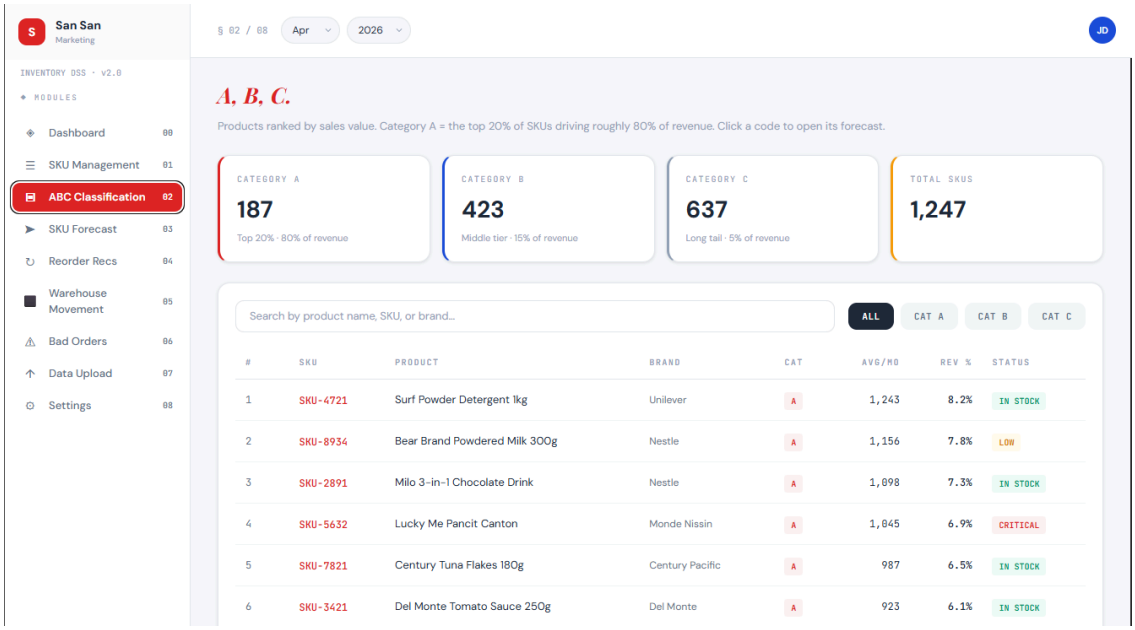
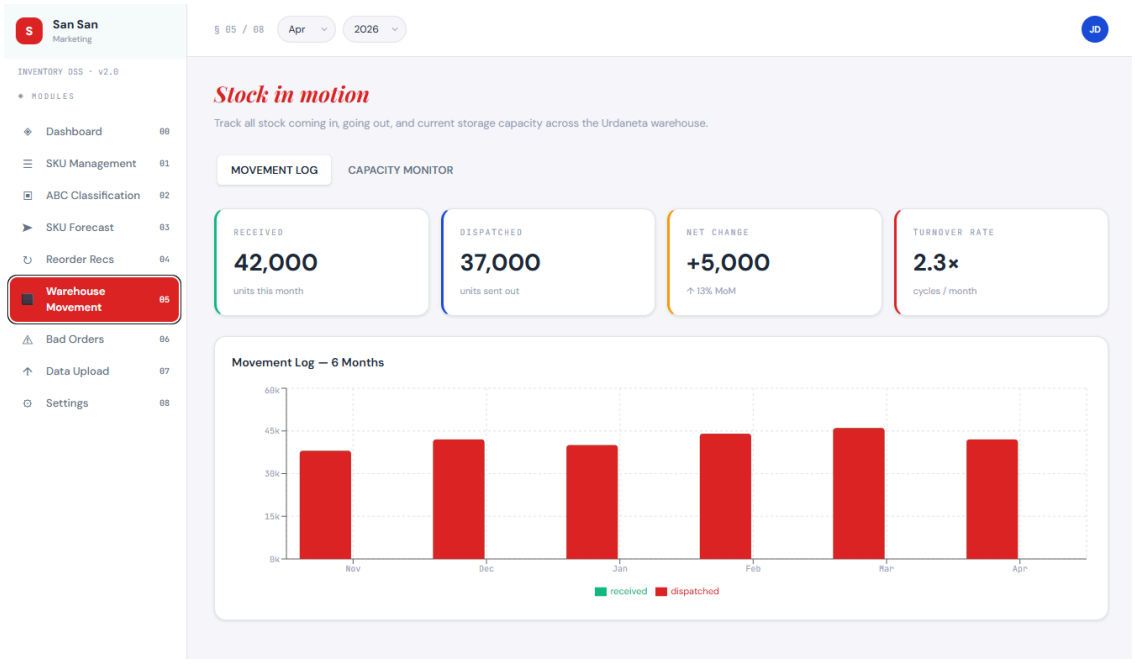


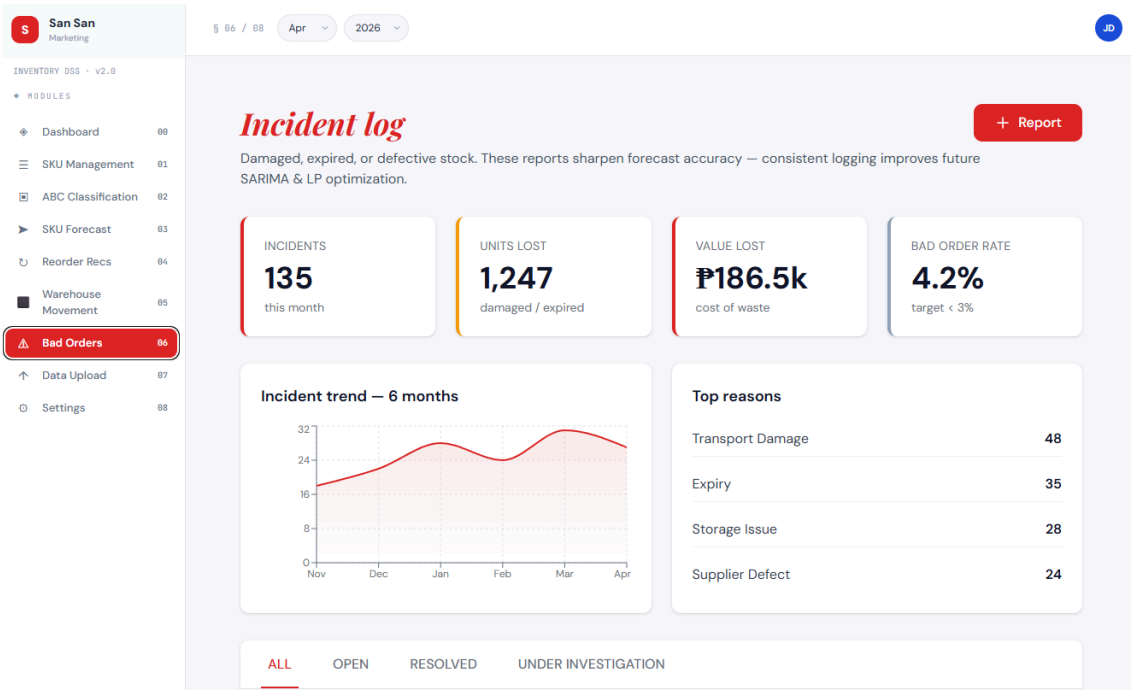
Figure 3.7.1 Dashboard Overview



3.7.2 ABC Classification Module



3.7.3 Warehouse Movement Module



3.7.4 Bad Orders Module